

Final Thesis Report

A Cross-Dataset based Zero-Day Intrusion Detection System by Integrating Siamese Network and Reinforcement Learning

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
October 2025

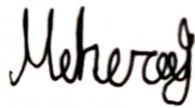
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Declaration

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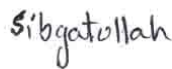
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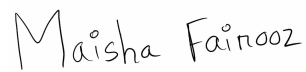
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Abstract

The threat of zero-day attack is looming over the Internet of Things (IoT) network security, which demands cognitive detection mechanisms that have the ability of sensing and acting on emerging malicious behavior. Conventional intrusion detection mechanisms with the capability of generalizing merely with given signatures of given attacks do not generalize to dynamic zero-day exploits within sophisticated IoT environments. The paper proposes a hybrid zero-day intrusion detection system using Siamese network-based anomaly correlation and reinforcement learning-based adaptive defense. The paper uses unsupervised machine learning classifiers over benchmark IoT datasets with the intention of detection of known attack types that will be compared to unknown anomalies using distance-based similarity analysis to detect possible zero-day attacks. To facilitate adaptability, a Proximal Policy Optimization (PPO) reinforcement learning-based agent dynamically adjusts the defense policy with continuous feedback and optimization. Experimental evaluations demonstrate 99.28% training accuracy, 99.07% accuracy in unknown attack detection, and 93.94% zero-day detection ratio, confirming the convergence and stability of the model on datasets. The system offers a proactive self-learning and extensible defense mechanism of IoT deployments by finding the right balance between precision and latency and false positives in the defense. This deep anomaly correlation with adaptive reinforcement learning is a firm base on which the next generation and autonomic cyber security solutions can take the reins as the zero-day threats keep changing their course.

Keywords: Zero-Day, Deep Q-Networks(DQN), False Positive Rates, Reinforcement learning, Anomaly Detection, Siamese Network, Proximal Policy Optimization(PPO).

Acknowledgement

We are thankful first of all to Almighty God for having given us strength, patience, and knowledge to finish this thesis. We are thankful again, secondly, to our supervisor, Dr. Muhammad Iqbal Hossain, for having been encouraging and giving useful advice guiding us to complete the thesis successfully. We are thankful also, thirdly, to our parents, whose kind patronization and blessings have motivated us to complete this thesis.

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Chapter 1

Introduction

1.1 Background

With the increasing number of people with the internet connection 24/7, cyber attacks are increasingly becoming more common and unidentified, which puts individuals, companies and even governments in danger. The one that is more insidious and threatening is the one that takes advantage of the issue of software and even hardware without the knowledge of the security personnel and is the zero-day attacks. These vulnerabilities leave developers with “zero days” to create and deploy a patch. Companies face financial losses, sensitive data violations and operational disruptions. Traditional detection systems based on attack signatures struggle to keep pace with mutating threats. Attackers frequently modify these patterns, making them challenging to identify and stop[10] [26]. To solve this, we need advanced and malleable detection techniques.

Anomaly-based systems, which focus on identifying behavior and offer an alternative but are often plagued with high false-positive rates. These false positives flood cybersecurity analysts with excessive alerts, making it harder for them to work efficiently; undermining the reliability of these systems in real-world use[20]. Nowadays, attackers employ sophisticated methods like code mutation, encryption, and anti-analysis mechanisms to evade detection. Identifying zero-day attacks becomes much more difficult with these advanced techniques[1] [12].

Reinforcement learning (RL) is also able to scan and process enormous volumes of data by seeking out concealed trends and patterns. PPO (Proximal Policy Optimization) and DQN (Deep Q-Networks) empower the systems in responding to unknown attack patterns; this is considered a limitation shared by all types of static pattern detection systems[28].

1.2 Rational of the Study or Motivation

Zero day attacks imply the weakness of the existing cybersecurity systems. Traditional strategies do not work without limitations as they are only based on the existing information and cannot be adjusted to new, unknown attacks. They are a reflection of their high demand for products which are lively and resourceful enough to match up the current dynamic front of cyber-attacks. The attackers use innovative approaches in order to circumvent the stationary defenses and this leads to the requirement of effective solutions[4].

In response to such problems, hybrid systems may be used. The Hybrid system is a combination of various successive models such as machine learning and reinforcement learning. We preferred reinforcement learning to provide the capacity of adapting and being dynamic in dealing with emerging attacks. This work gives the method of recognizing the zeroday attacks more efficiently and more precisely.

The value of this study is that it relates the latest developments in the area of artificial intelligence with the practical problems of a cybersecurity expert. The most vexing issues that continue to bother existing security situations are addressed in our proposed framework- the frustrating false alarms that flood the analysts and more alarmingly, the threats that sneak in without being noticed. In the end, the main focus is to improve and develop the detection capabilities and construct more secure cybersecurity defenses that can be effectively trusted and relied on by security teams.

1.3 Problem Statement

Zero-day attacks are very dangerous since they take advantage of the invisible weaknesses in software; hence attackers can gain access into networks before the software is patched or security updates have been implemented. These are popularly referred to as "zero-day attacks," which describe how these types of attacks target any unidentified weakness in software and systems. These attacks are much more dangerous since they can't be detected by typical security systems based on the known attack patterns. Zero-day attacks are becoming more widespread and have a greater impact with computer systems becoming more sophisticated and more devices connected to the internet. It causes process issues, financial loss, and data leakage[18].

Detection and prevention of such attacks in real time is the most challenging job, however, as they mostly adopt routine behaviour and hence are hard to detect. Current signature-based systems cannot deal with the growing and changing nature of zero-day threats. Zero-day attacks are most threatening because they attack unknown vulnerabilities and pose threats to individuals and organisations alike. These attacks have the potential to bring key infrastructure down, cause colossal losses to economies and reveal massive volumes of sensitive information. They frequently get through both conventional security measures since they take advantage of un-

charted weaknesses and, as such, zero-day threats present one of the most difficult problems in cybersecurity[30] [1]. The solutions to these attacks build confidence in digital systems by protecting sensitive information and reducing how often tasks are disrupted. Locating and mitigating zero-day attacks becomes a major priority toward good cybersecurity, ensuring the security of data and minimizing the threats of dynamic and ever-evolving cybersecurity challenges.

1.4 Objective

Primary objective of our research work is to conceive and implement a multilayer hybrid model of artificial intelligence for zero-day attack identification with highest accuracy and a minimum number of false positive outcomes. This model combines unsupervised machine learning frameworks, Siamese neural networks, and reinforcement learning algorithms to form an adaptive cybersecurity system that can recognize previously unknown attack patterns in an IoT network.

- Demonstrate robust performance across multiple IoT datasets (CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024).
- Develop a three-layer detection framework combining unsupervised anomaly detection, cross-dataset siamese analysis, and reinforcement learning based adaptive response. Run 5 unsupervised machine learning algorithm including PCA-Based Anomaly Detection, Isolation Forest, Standard Autoencoder, One class SVM, K-means to check which one performs best.
- Design a Siamese network that makes comparisons across-dataset anomalies with different distance metrics like the Euclidean, Manhattan and Cosine similarity metrics.
- Conduct comparative research on the variety of RL algorithms such as PPO, DQN, SAC and create adaptive reward functions that would trade off threat detection performance with operational performance.
- Test how well the system identifies zero-day attacks unseen during data seen during training. Present the overall monitoring of the performance in terms of convergence speed, resource consumption, and detection indicators.

1.5 Methodology in Brief

The method is based on two massive datasets of traffic on IoT that do not include any manual labels. The system detects common and previously unknown attack patterns by default by working with unlabeled data, no supervised training or human annotation is needed. It is designed as a combination of an innovative anomaly detection pipeline and a reinforcement-based learning defense, which constantly adjusts and provides real-time defensive capabilities that are adaptable to the changing threats.

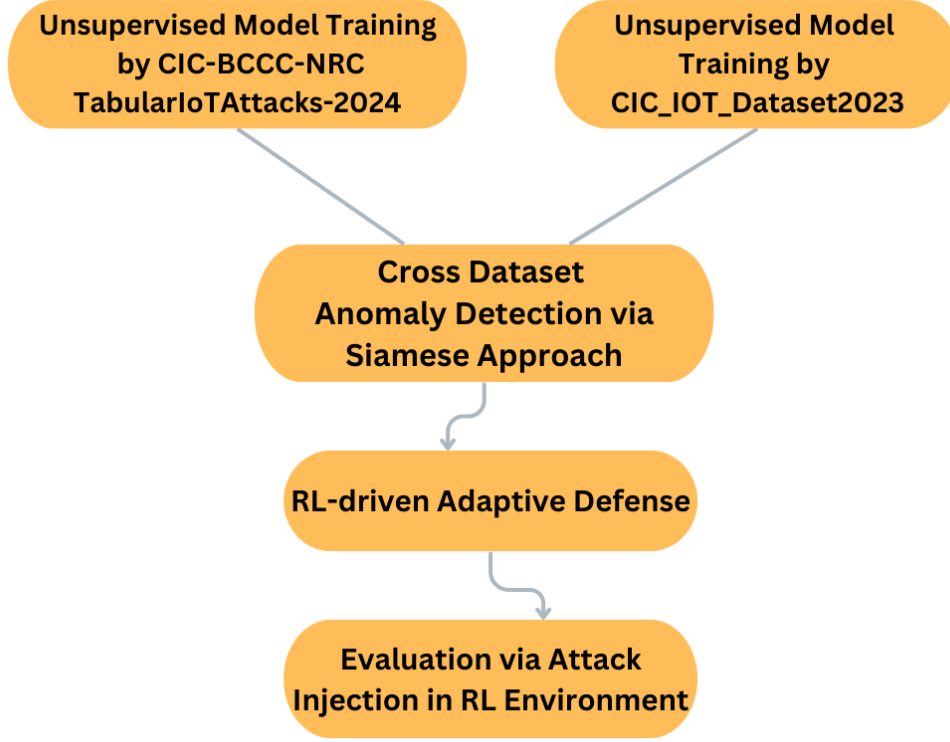


Figure 1.1: Work Flow

We firstly train anomaly detectors on each dataset independently with their shared features. Then we compare the identified anomalies in datasets with various distance measures in order to cull out uncommon occurrences. Lastly, the remaining anomalies supply three reinforcement learning agents, which learn and perfect defensive strategies independently, forming a self-optimizing security machine, which keeps pace with the emerging cyber threats. The entire system is then tested by considering the response of the system to different injected attack situations.

Workflow Explained

Model Training by Datasets: Unsupervised anomaly detection models are trained individually on two large-scale IoT attack datasets. One is CIC BCCC NRC-TabularIoTAttacks 2024 and the other is CIC IoT Dataset 2023 with only semantically matched shared features retained. The training involves a number of heterogeneous algorithms, including PCA-based detection, Isolation Forest, Autoencoder, One-Class SVM, and K-means, to enable the models to learn a baseline of network traffic pattern behaviors without labeled data.

Cross-Dataset Anomaly Detection via Siamese Approach: Anomalies extracted from both datasets are subjected to a Siamese-style comparison with three distance measures: Euclidean, Manhattan, and Cosine distances. Here, the best threshold possible is selected for identifying dissimilar and rare attack patterns between datasets for each metric. And based on AUC score, spread and F1 score, the best metric is to be selected. This measure is selected for ultimate anomaly extraction based on this comparison for optimum detection fidelity.

RL-Driven Adaptive Defense: Filtered anomalies serve as training data for a reinforcement learning agent based on an Actor-Critic architecture and a value based method. We have evaluated three algorithms which are Proximal Policy Optimization (PPO), Deep Q-Network(DQN) and a SAC model. The RL agent is trained only on attack data with the goal to learn adaptive defense actions such as selective blocking and alerting.

Evaluation via Attack Injection in the RL Environment: The PPO learning model is tested with controlled injection of two different data sets: (1) the training set, to ensure the agent’s attack-response behavior is retained after learning, and (2) an entirely unknown set, to ensure its zero-day attack detection capacity. It guarantees that the suggested model below is capable of obtaining high precision for both phases of discovery, adaptation, and resistance for both unknown and known malicious patterns of a traffic.

To conclude, by integrating unsupervised anomaly detection, strict cross-dataset anomaly validation, and reinforcement learning-driven response mechanisms, the presented pipeline in figure 1.1 moves proactive defense forward for zero-day attack detection and mitigation in IoT networks.

1.6 Scopes and Challenges

Detection and prevention of zero-day attacks is one of the most complicated problems in the existing realm of cybersecurity as it requires a highly technical answer, not only in the sense of accuracy of the detection, but also with regard to the operations of deployment. The hybrid detection solution presented here has tremendous potential of enhancing this field by creating better capability in detection performance and new abilities of using the Siamese networks in cross-dataset pattern matching to detect more precisely the new attack vectors that have diverged out of the traditional signatures. The design of the multi-layered architecture has the potential to make the results more accurate, by adding algorithms in autoencoder-based anomaly detection, PCA dimensionality reduction, and reinforcement learning to the design, in an attempt to reduce the false positive (FP) and false negative (FN) results compared to traditional detection algorithms. The reinforcement learning will be to support adaptive responding in real time so that the system will learn and adapt automatically to new patterns of threat with no human intervention/or signature updates. There are different types of challenges that must be encountered during the implementation to install and maintain the system effectiveness. The system requires a lot of computing abilities i.e. high infrastructure cost which might be prohibitive to the small organizations. The quality of data is also an issue of serious concern, since the usefulness of machine learning models directly relies on the access to clean and representative training sets, noisy or biased data can easily make the system performance significantly worse. The problem of scalability is particularly problematic when using AI-based security systems in large network-based enterprises, where a high level of data processing is required to be maintained, yet rapid response time requires the use of highly improved algorithms. Regular model

updates and retraining cycles are also necessary to maintain detection pace with new attack patterns because the threat environment is constantly evolving. Additionally, other ethical aspects like privacy protection, explainability of algorithms, and proper deployment of AI should be incorporated cautiously by execution in a bid to meet regulatory standards and trust of stakeholders.

Chapter 2

Literature Review

2.1 Preliminaries

Reinforcement Learning (RL) : Reinforcement learning (RL) is a specific type of machine learning that conditions an agent to make a decision by engaging with the world. Deep learning is combined with Q-learning to develop techniques such as Deep Q-Networks (DQN) which are particularly effective in handling zero-day attacks by learning a policy to maximize rewards over time. One of the reinforcement learning algorithms Proximal Policy Optimization (PPO) is employed for its stability and efficiency in training.

Anomaly Detection: Anomaly detection is a technique to detect out-of-pattern or out-of-behavior data in data sets that are other than normal patterns. Such anomalies can be indicative of errors, fraud, or attempts at hacking. The segregation of normal and abnormal data is performed using statistical techniques, machine learning, or deep learning models. Anomaly detection has widespread use in cybersecurity, finance, healthcare, and network traffic analysis.

Intrusion Detection and Prevention Systems (IDPS): An intrusion detection system (IDS) monitors system or network activity with the aim to detect unauthorised activity or policy violations. When a threat is suspected, they inform system administrators with immediate blocking or mitigation of the coming attack in order to ensure integrity within the network.

PCA-Based Anomaly Detection: Principal Component Analysis (PCA) is a concept of dimension reduction that is unsupervised, projecting data onto a space of fewer dimensions, but not shrinking the largest variance. With anomaly detection, PCA can determine normal patterns after learning the principal components which carry most of the data variance. The error in reconstruction anomalies - difference between the original data points and their reconstructed versions after the reduction of the dimensions - measures anomalies. High reconstruction errors on data are supposed to be anomalous as they differ so much with the normal patterns that are learned. Anomaly detection with PCA especially Identifying outliers in

high-dimensional data Data computationally-efficient PCA is ideal in computing on large-scale data.

Isolation Forest: Isolation Forest is an anomaly detection technique, which isolates anomalies by choosing features randomly and dividing values between values of sampled features and their minimum and maximum values. The algorithm assembles a collection of isolation trees in which anomalies are sheltered nearer the tree basis, needing a fewer split than normal examples. The key concept here is that anomalies are rare and distinct hence simpler to pick out of a set of regular data points. The anomaly score of a data point consists of calculating an average path length across an entire forest of trees. The isolation forest is relatively effective in finding global anomalies and it operates effectively with high-dimensional data without prior knowledge of usual patterns of behavior.

Standard Autoencoder: The simplest type of an artificial neural network, which trains to learn useful representations of the input data, is a standard autoencoder. It is comprised of two major constituents: an encoder, to compress information into a latent space of a lower dimension, and a decoder, through which the original input data is reconstructable based on a compressed format. The encoder and decoder are normally composed of fully connected layers that are organized in a symmetrical manner. The concept during training is to minimize the error between the primary input and the reconstruction of it. The normative autoencoders predominantly learn how to ignore the least significant features of information in a data set by removing redundancies and noises. They can be used to reduce dimensions and to detect anomalies since the samples with the highest reconstruction errors are usually outliers. The standard autoencoder is suitable for extracting nonlinear, complex data associations by employing no other regularizations or noise schemes of improved variants.

One-Class SVM: One-Class Support Vector Machine is an unsupervised learner, which finds a hyperplane of a high-dimensional feature space, which separates normal data and the origin, usually by means of a kernel function. The algorithm learns solely with normal data, and it is trained to place a border which would cover most of the normal cases and would occupy as little space as possible. The points that are beyond this learned tip are defined as anomalies. One-class SVM is applicable to handle noisy data, and also for high-dimensional space, and is hence suitable to employ if only normal data for training is available.

K-means: The K-means clustering is an unsupervised clustering method which partitions the data into k groups based on a sum of squares minimization criterion. K-means finds normal patterns in order to detect anomalies through formation of clusters of similar data points. Anomalies are identified depending on how far they are with respect to cluster centers - data points far away relative to all the cluster centers are anomalous or they fall in a very small cluster. The iterative operation is repeated wherein data points are assigned to nearest centroid and updated clusters until centroid convergence. K-means detects anomalies quickly and effectively when normal data is separated into distinct and well separated clusters, it can however be difficult to pick up clusters of different density, or irregular shapes.

Actor-critic architecture : Actor-critic structure reinforcement learning model is categorized into value-based and policy-based models by sharing their merits. It consists of two constituents. The Critic learns value functions and critiques the actions. The Actor learns the policy by choosing the actions for a certain state. The Critic provides the Actor with temporal-difference (TD) error feedback to help with changes to the policy. The design advances learning by increasing convergence over purely value-based methods and decreasing variance over purely policy gradient methods. Advanced algorithms such as PPO, A3C and DDPG extensively implement actor-critic models for effective learning in complex, continuous spaces.

Proximal Policy Optimization (PPO) : One of such algorithms of reinforcement learning is Proximal Policy Optimization, or PPO for short. It helps stabilize training and optimize for highest efficiency across samples. By restricting the change in policy compared to the clipped replacement objective, this policy gradient method prevents big updates that would lead to instability in learning. PPO maximizes actions continuously and optimizes policy within an ongoing "proximal" neighborhood of the current policy to realize the exploration-exploitation trade-off. The PPO approach performs similarly or better on many types of scenarios, but it is simpler to implement and cheaper compared to earlier algorithms like Trust Region Policy Optimization (TRPO). It is used widely in robotics, games, and anomaly detection scenarios where stability and persistent learning are required to handle dynamic and uncertain environments.

Deep Q-Network : DQNs combine Q-learning and deep neural networks, which are able to handle complex, high-dimensional worlds. Although classical Q-learning is burdened by Large state-action spaces, DQNs neural networks approximate Q-value functionality and learn successful tactics by raw images input. DQNs learning is stabilized with the use of target networks, which provide fixed target Q-values for updates, and experience replay, which holds and recycles past experiences to break associations.

Siamese Network : A Siamese network is composed of two neural networks that are identical with the same parameters and weights to learn the similarity between input samples. It accepts two inputs at a time and compares their feature integration to predict their similarity. The network is best suited for comparing pairwise tasks or one-shot learning with sparse information since it learns to score similarities by comparing feature embedded data.

Soft Actor-Critic (SAC): Soft Actor-Critic (SAC) is an off-policy reinforcement learning technique which finds a stochastic policy of maximization of a trade-off between expected reward and entropy. It runs actor and twin critic networks to predict the policy and value functions with accuracy and eliminates bias of overestimation. SAC has entropy regularization that promotes the wide range of action choice, resulting in more stable and sample-efficient learning. It fits extremely well to continuous action spaces and high-dimensional problems and is thus effective to use in real-time cybersecurity threat detection. The algorithm uses automated entropy tuning and provides experience replay to enhance training efficiency, which makes it robust and converges quickly.

2.2 Review of Existing Research

Brinkley et al. [22](2024), in their research, utilized machine learning methods like Random Forest, SVM, and Neural Networks that were able to detect new, unseen before malicious activity. The researchers employed a sample of 2,850 samples taken in the public cybersecurity repositories in their model. They did extensive data cleaning, data normalization and data segmentation to get the data ready to be trained. Extraction techniques were used to extract features of normal and abnormal network traffic in order to extract complex behavior patterns. The neural network has been the most accurate in the end analysis with a score of 92.4% followed by the random forest model with 91.2%.

Satam et al. [18](2023) thoroughly reviews different solutions, such as anomaly detection, machine learning (ML), and hybrid models, in the counteraction of zero-day attacks. They especially discuss the use of threat intelligence applications like VirusTotal and MetaDefender, which provide external validation to enhance the results of reinforcement learning (RL) applications. Such useful attacks as SQL Injection and Remote Access Trojan are also included in the paper to demonstrate the behavior of the attacker. Network traffic monitoring is highly related to these simulations.

Pitre et al. [10](2022) suggest a hybrid two-stage intrusion detection system (IDS). They are based on a combination of feature selection and fine-tuning methods to enhance anomaly detection. At the initial stage they use the logistic regression, and at the second stage they fine-tune the model and make it more specific on differentiating false positives and true positives. The paper also discusses the various optimizers, including Adam and RMSProp, to improve the speed and accuracy of the classification.

Zaki et al. [30](2024) create an IDS to detect zero-day attacks in the IoT network with the help of ensemble machine learning. To select features, they jointly use sparse random projection and a composite model of classifiers that use k-Nearest Neighbors (KNN), XGBoost and Stochastic Gradient Descent (SGD). This is an efficient method of reducing data dimensionality and at the same time retaining the most valuable information to train. The model is based on the CIC-DDoS2019 and CICIDS2017 datasets and the model scores 99.91% accuracy and with a detection time of 0.22 seconds can be applied in real-time. The paper's focus on applying GridSearchCV to optimize each classifier's hyperparameters guarantees improved performance over earlier methods.

Kareem et al. [20](2024) in their paper used Neural Transformers for Zero-Day Detection. The transformer model processes sequences of network packets (size of the packet, protocol type, source, and destination addresses) to learn patterns that are typical of normal traffic and those that are unusual. They used public datasets like CICIDS 2017 to provide diverse examples of attack patterns. They claimed that they attained an accuracy of 96% and precision of 94% in the transformer model, which is much higher than logistic regression, SVM, and random forest. Also, the average latency per packet was 25 milliseconds, which is significantly lower in

comparison to other deep learning models tested.

Abbas-Escribano and Debar [12](2023) proposed an improved high interaction honeypot with an emphasis on having an increased level of interaction with attackers for purposes of detection and analysis of attacks while maintaining system integrity. This improved model consists of virtualized architecture elements positioned at different network levels such as entry, intermediary, and internal. The model derived its basis from the MITRE ATT&CK taxonomy for mapping adversary tactics and techniques. These include brute-force SSH and RDP service attacks, as well as broad scanning activity. These were identified by the recording of this improved honeypot technique for the time span of over 17 days, during which 1.5 million events were recorded. The most prominent finding of this technique is that the attackers were not allowed to successfully penetrate the honeypot's deepest layers, thus the model logged the behavior of these attackers very well, providing valuable Indicators of Compromise (IoCs).

Diloglu [4](2022) suggested zero-day attacks detection through deep learning models in the systems of connection, and his model outperformed the traditional detection algorithms for all measures of evaluation such as F-score, accuracy, and detection time. The proposed methods were trained and tested using KDD '99, CIC-IDS2017, NSL-KDD and UNSW-NB15 datasets. The approach was grounded on supervised and unsupervised deep learning algorithms. It included artificial neural networks and autoencoders with batch normalization with the objective of improving the attack detection capabilities. High accuracy was achieved by supervised models, and from the findings, the CIC-IDS-2017 dataset had a 97% accuracy rate, while unsupervised models were poor in classification in older datasets.

Balamurugan, M. [21](2024) attempted to show the extent to which honeypot utilization and artificial intelligence reduces detection and mitigation in zero-day attacks in their paper. This work also emphasizes the privileges of AI and honeypot systems in detecting unknown attacks and preventing lateral movement of threats in computer networks. For this work, supervised and unsupervised machine learning models are being used. The name of the datasets used for this research paper is UNSW-NB15, CICIDS2017. In the case of AI-enhanced honeypots, the detection rate is 92%, whereas there is a 75% detection rate in the traditional systems. Average response time came down from 5% to 2% compared to the conventional systems. The false positive rates have been decreased from 10% to 3%, whereas the false negative rate went down from 8% to 2%.

Igugu, A. [24](2024) in his article talks about the use of machine learning and artificial intelligence methods in cloud infrastructures to identify zero-day-attack detection. Here, Catboost, XGboost, application of neural networks, Support Vector Machine and collective learning methods are utilized. In this article, the AWS-CIC 2018 dataset has been used. Ensemble learning models regularly excelled other algorithms with a success rate of up to 99% for known attacks.

In their academic research work, Nkongolo et al. [9](2022) took the ambitious and challenging work of developing a new cloud-based methodology for zero-day attack detection, notoriously challenging to detect, and developed it in a novel manner.

For developing such an innovative model, they utilized the rich UGRansome1819 dataset, a strong basis for such work. In this particular piece of work, in a novel genetic algorithm, with just a single objective in mind, i.e., to maximize and enhance performance measures for multiple types of machine learning algorithms, they used a particular algorithm, i.e., commonly utilized and popular algorithms like the Support Vector Machine, Naive Bayes, and Random Forest, and these algorithms have high regard and usage in the machine learning field, an indication of their usefulness and usage in overcoming challenging computational issues. The cloud services used for building and validation include the S3 bucket and SageMaker by Amazon Web Services. The research shows that accuracy of 99.6% is achieved by the Random Forest model and it outperforms other models on CAIDA and UNSWNB-15 datasets. The research showed some key results. It found that using ensemble learning boosts how well weak classifiers like Support Vector Machine and Naive Bayes perform when paired with the Genetic Algorithm. Genetic Algorithm was also useful in feature selection and in reducing computational time and overfitting by eliminating most of the less important attributes.

Wahed et al. [38] (2025) proposed an AI-enhanced IDS that employs advanced machine learning deep learning models, particularly RNN, LSTM, and GRU. These models have the ability to analyze sequential data, in this case network traffic data. No particular datasets were used but the training involves labeled network traffic data; approximately 2-3 million records per attack type. The architecture is capable of processing real-time big data with predictive analysis, significantly reducing detection time whilst keeping false positive rates (FPR) below 8%. The RNN was trained to recognize patterns of potential threats. To further enhance detection accuracy and improved optimization the architecture integrates Random Forest after rigorous hyperparameter tuning. The architecture also employs natural language processing (NLP) techniques for examining security reports and bulletins. This reconstructs data, meaningful to the system, from unstructured textual data. The architecture is deployed into a real time IDS for live detection and automatic trigger responses like disconnecting suspicious sessions or alerting security teams. Overall, the architecture proves to be robust and scalable while providing proactive solutions.

Hairab et al [7](2022) clarifies utilizing CNNs classifier to identify unknown malicious attack traffic. Various techniques like L1 and L2 have been used to avoid overfitting and control the complexity of the classifier. Various types of attack data like Service Scanning, Data Theft, Key Logging, OS Fingerprinting, DoS and DDoS are present in this dataset. They work better on all the metric evaluations when applying regularization methods as opposed to traditional CNN. They trained, tested, and cross-checked classifiers in the aforementioned cases on the Bot-IoT dataset. This dataset holds various attack data such as Service Scanning, Data Theft, Key Logging, OS Fingerprinting, DoS and DDoS. They achieve their best 99.98% accuracy with CNN L2.

Mbona et al [8] (2022) demonstrates that semi-supervised ML methods are efficient for zero-day attacks detection when important features are selected optimally.

They report the best performance by applying one-class support vector machines to zero-day attack detection with an F1 score of 85%. Four datasets were utilized for this article to examine zero-day network intrusion attacks such as UNSW-NB15, CICDDOS2019, OTINTRUSION2020, CIRA-CIC-DOHBRW-2020. They made the opinion that such features, which were characterized mostly significantly different between zero-day network traffic and normal network traffic. This could be determined using the law of anomalous numbers, or Benford law.

Kandhro et al. [15](2023) in their paper they've written about the use of CNNs, RNNs, and autoencoders, to detect malicious intrusions in IoT-driven environments. Their study concentrated on how well traditional intrusion detection systems detect online threats, particularly zero-day attacks. They trained algorithms using NSL-KDD and UNSW-NB15 datasets. They were able to classify both common and uncommon network threats and get high accuracy. It uses real-time classification pipelines. However, the absence of a specific industrial environment and testing against honeypot logs limit this research's potential. We fill the gap by combining anomaly detection with real flows produced by honeypots.

Faldi et al. [13] (2023) implemented a honeypot-based server security system to analyse how different types of cyberattacks interact with network defense systems. Honeypots are particularly effective at detecting application-layer attacks, and can provide valuable early-warning data when we are using signature-based IDS with it. They placed honeypots to monitor attacks such as GoldenEye, LOIC and Slowloris by using pfSense and Snort IDS in a controlled lab environment. A key strength of their setup was its simplicity and real-time alert capability but there is a major problem. That is, their system can't predict new attack patterns, especially those not previously found. Here, we are referring to not just logging attacker behavior through honeypots, but also using machine learning models to identify new zero-day attacks by anomaly modeling and pattern similarity.

Ali et al [3] (2022) highlighted methods of artificial intelligence for the identification of zero days attack, which are difficult in comparison to conventional detection techniques based on known vulnerabilities. It discusses several attack scenarios with datasets including ICS Gas Pipeline, ISOT, ISCX, NSL-KDD, CIDD, CICIDS2017, SPMD, Enron Spam, UNSW-NB15, Kaggle Microsoft Malware, and SWaT. Decision Trees, Random Forests, MLP-IDS, Bloom Filter, LSTM, HML-IDS, Kalman Filter, Zero-Shot Learning, SVDD, Autoencoder, CNN, Reinforcement Learning, DNN, tDCGAN, WAVED, Grey Wolf Optimiser, Stacker, and MSALSTM-CNN are some of the algorithms considered in this work. Stacker (99.97% on CICIDS2017) and tDCGAN (95.74% on malware datasets) are top performers when evaluated. Evaluation is based on accuracy, precision, recall, and F1-score.

Topçu et al [19] (2023) provide a novel way of finding zero-day cyber attacks by mining Twitter posts. The approach is based on actual time, public tweets to alert people about problems before these become widespread. The authors have extracted the tweets with the terms zero-day, zeroday, and 0-day. That text is then cleaned,

sorted, and mined with a mix of NLTK and TensorFlow tools. NLTK tackles the routine tokenization and basic security checks while TensorFlow quickly turns each tweet into tidy, small vectors that lift overall accuracy. Its efficacy compared to conventional methods is testified by its 80% detection rate with TensorFlow and 60% detection rate with NLTK.

Guo [14] (2022) discusses critically methods that utilize machine learning to identify zero-day attacks, which take advantage of newly found software vulnerabilities and mostly evade conventional security measures. Several datasets have been reviewed in the paper, such as NSL-KDD, CIC-IDS2017, CSE-CIC-IDS2018, and private datasets. The study notes that feature values change widely and that realistic zero-day samples are almost absent. The author reviews a mix of methods, naming supervised and hybrid tools like Random Forest and Support Vector Machines, plus unsupervised options such as One-Class SVM and autoencoders, and finally transfer-learning approaches like manifold alignment and spectral transformation.

Jurečková et al. [25] (2024) developed a method that uses both classification and clustering techniques to handle malware and in the process was able to isolate zero-day malware. They use the EMBER dataset which contains around 400,000 labeled malware samples. At first, static analysis is used to extract features without executing the malware samples. These features and static attributes are then used to train a multilayer perceptron (MLP) classifier, a kind of feed-forward neural network. This is able to assign incoming samples to known malware families. For samples that cannot be successfully classified by the MLP, they are clustered into new groups using self-organizing maps (SOMs), a sort of neural network that is unsupervised. Finally, the newly formed groups are analyzed and if conclusive are integrated into the overall malware detection framework and so that later, newly identified variants are easily classifiable.

Roopak et al. [27] propose an unsupervised system for intrusion detection to identify zero-day distributed denial of service (DDoS) attacks in IoT networks without any previous ideas of the patterns of attack. The model proposed by them uses Gaussian Random Projection (GRP) for the reduction of feature space, which effectively lowers the feature space from 87 dimensions to 25 dimensions while still maintaining important structural relationships. The suggested ensemble method integrates three unsupervised classifiers—K-means, Gaussian Mixture Model (GMM), and Stochastic Gradient Descent-based One-Class SVM (SGD-OCSVM)—through the application of a hard voting technique to enhance classification robustness. The system was tested on the CIC-DDoS2019 dataset and got accuracy of 94.5% and F1-score of 94.3%, which outperforms some state-of-the-art unsupervised methods. Roopak et al. note that the model can effectively handle high-dimensional, unlabeled data with efficiency, making the model amenable to real-time detection of new IoT-based attacks.

ML and DL approaches have recently become popular in dealing with Zero-day threats. Dahal et al. [23] (2024) in their paper also use ML and DL, specifically multilayer perceptrons (MLP), and Explainable AI (XAI) for extensive interpretability. They utilize the KDD99 dataset, which is a popular intrusion detection system dataset that contains about 4.9 million data samples and 41 various features classes into 23 classes. In order to counter class imbalance, the authors merged these 23 classes into 4 general categories: These are distributed denial of service (DDoS), unauthorized access, probe attacks, and normal. This data is used to train four different multilayer perceptrons, which include baseline-which was the standard model trained on original data, weighted-trained with class weights, truncated-a smaller model with fewer parameters, neurons and layers to improve training speed, and finally weighted truncated models-combining the truncated architecture with class weights to handle class imbalance.

2.3 Summary of Key Findings

A number of studies over the recent years have also found some significant limitations and the future outlook of zero-day attack detection models. Brinkley et al.[22](2024) pointed out issues with resource optimization and scalability and pointed out that their Neural Network model consumed more CPU and memory during testing and added latency in detection, indicating lightweight Neural Networks or ensemble methods to reduce complexity and adding adaptive learning to real-time ransomware activity detection. Satam et al.[18](2023) identified design and practical failure, in particular, with honeypots, and the absence of real-time adaptation and dynamic response to zero-day attacks despite useful underlying RL-based knowledge. Pitre et al.[10](2022) enhanced false positive reduction with low false negatives but achieved only on the CIC-IDS2018 dataset. Zaki et al. [30](2024) focused on ensemble techniques to detect zero-days, but this study was limited in its applicability to general-purpose networks and real-world scenarios, as it was an IoT-specific analysis. Computational complexity was found to be a limitation of the transformer-based models, with quantization and knowledge distillation being suggested by Kareem et al. [20](2024). Abbas-Escribano and Debar [12](2023) observed a short exposure to honeypots and low interest by advanced attackers, suggesting a longer deployment and improved data collection. According to Diloglu [4](2022), it was recommended to pay attention to real-time data and combine deep learning with old algorithms, such as Bayesian optimization, to achieve high accuracy and efficiency. Balamurugan [21](2024) emphasized that AI is susceptible to adversarial attacks and difficulties in minimizing false negatives, and claims that the combination of IoT with blockchain and federated learning can increase detection and privacy. Igugu [24](2024) observed that, although ensemble ML models can have a lot of potential in the zero-day attack counter, they were still in the development phase to be adaptive and effective in defending the cloud environment and emphasized less on non-ML or other cloud defenses. Nkongolo et al. [9](2022) cited the instability of classifiers and lack of data, and suggested more data collection and deeper networks. Wahed et al. [38] (2025) achieved high accuracy (98.5%) with RNN, LSTM, GRU models but indicated a need to optimize and retrain the model

continuously and dependent on the quality of training data. Hairab et al. [7] (2022) noted that ML/DL overfitting occurs because of the complexity of such models and insufficient data dispersion, indicating that time could be optimized better to train the CNN and generate various Botnet attack scenarios. Mbona et al. [8] (2022) discovered that Benford law was restricted to positive real numbers and suggested combining several feature selection techniques and ML classifiers. The study conducted by Kandhro et al. [15] showed deep learning to be effective in detecting zero-days in the IoT, however, without industry validation and honeypot integration, as our study provides. Faldi et al.[13](2023) demonstrated the usefulness of honeypots using known attacks and limited to new patterns, suggesting a better ML anomaly detector to expand zero-day detection without compromising real-time notifications. The researchers pointed out shortcomings like old datasets, high false positives / false negatives, bias in datasets, and slow response time (Ali et al.[3](2022)), suggesting mixed-method models, new metrics, and the implementation of IoT devices to ensure better zero-day detection. Topcu et al. [19] (2023) pointed out the excessive use of key-word searching, scaling limitations, and manual processing in social media detection, suggesting adaptive self-learning models and extended real-time detection. Guo [14](2022) pointed out unequal detection accuracy because of dataset and metric constraints, suggesting the use of domain knowledge, reinforcement learning, honeypots, and standard reference collections to detect better zero-day attacks. Jurecekova et al.[25](2024) created an online processing model with 97.21% accuracy with clustering of uncertain samples. Recognizing the merits of zero-day DDoS detection, Roopak et al.[27] emphasized the complexity of the ensembles and the computational cost, suggesting lightweight clustering and adaptive learning as an option to improve scalability and adaptation to additional attacks. It was revealed that truncated models with lower parameters performed worse than baselines but could better handle class imbalance at the cost of lowering the model (Dahal et al.[23] (2024), highest accuracy at 99.62%).

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

Functional Requirements

Doing preprocessing requires correction of missing values, elimination of infinite values and data normalization. Data integrity needs to be upheld but appropriate scaling and noise reduction be done. The standard autoencoder where it is used to detect anomalies based on foundations and Principal Component Analysis (PCA) where it is used to reduce dimensions and extract latent anomaly features should be well incorporated in the model. The system will also use a Siamese network in similarity checking of datasets. The siamese network should be able to appropriately compute distance measures (Euclidean, Manhattan, and Cosine) and compare similarity between identified anomalies, select the ones that were actually new (zero-day) threats. The suggested design should use adaptive learning by implementing RL algorithms (PPO, DQN, SAC) and improving detection policies. The Reinforcement Learning (RL) component needs to operate in an agent-environment system. The model will have to communicate with the network traffic data in the form of PCA, Autoencoder, Siamese similarity scores and constantly revise his decision policy to fit changing attack patterns. The system should be able to produce automatically performance measures such as accuracy, precision, recall, F1-Score, AUC-ROC and RL measures such as convergence rate, resource usage, and detection latency. Each layer must export all the anomalies detected to individual CSV files with timestamps, classification context, variance, IAT to further analysis. The framework should be able to partition datasets to avoid training any unseen attacks so that to reproduce the actual zero-day conditions during the testing stage.

Non-Functional Requirements

Theoretically it should be scaleable in design. It must also be able to work with the IoT networks of any traffic capacity with insignificant performance loss. It should be end-to-end encrypted, possess access control and conform to pertinent standards (e.g. GDPR) and data-protection and integrity controls. The system should support version control and modular codebase to enable it to support debugging when it is needed, performance optimization and functionality extension. The system should issue analytical reporting on a periodic basis where they can be interpreted and read by cybersecurity researchers and analysts. The system should be capable of being deployed on several environments (e.g. cloud servers, on edge devices) without much reconfigurations. Lastly, the system must be operational at all times with minimal downtimes.

Possible Constraints

The constraints of proposed IDS occur in multiple core domains, the first of which is the constraints of the data: the work of the system directly depends on the availability and quality of benchmark sets like CIC-IoT2023 and CIC-BCC-NRC2024. Although they are open-source and full, they are not able to realize the variety of zero-day attacks in the real world, and incomplete labels, noisy features, and dynamic attack behavior bring limitations to the trainability process and the test. The time constraints are another important consideration as far as the fact that reinforcement learning is more computationally costly than the traditional ML based IDS schemes. It causes slow trainability and convergence times, and is slower than signature-based IDS or shallow learning-based IDS in field deployable scenarios, but once learned, the inferences speed should be the same, though real-time response may be problematic subject to the workload and scale of the system, but still, a major drawback in large-scale IoT application contexts, where large numbers of hardware/TPUs/GPU nodes are deployed, large CPU/GPU cycles in the process are required even of simple-inference of large installations even of even smaller scale, which in turn limits deployability. Lastly, data imbalance is also a limitation. Zero-day or rare attack samples are far fewer compared to benign/normal traffic or known attack variants, hence biased learning where the IDS tends to prefer benign prediction and raise False Negative risk, leading to failure to capture new zero-days, hence easy compromise of network by new zero-days, yet proposed model manages it by an integrated combination of modeling of anomaly-detection schemes and reinforcement learning-based shaping of reward, whereby autoencoder and PCA learn compact normal network behavior representations, instead of based on rich attack label, whereby deviations are flagged as potential zero-days without requiring balancing of train data.

3.2 Societal Impact

The social value of planning and deployment of a zero-day intrusion detection system (IDS) integrating Siamese networks with systems employing reinforcement learning, like Proximal Policy Optimisation (PPO), is significant. In the era of rapid digitalisation, where zero-day attacks impact people, companies, organisations, states, and critical infrastructure, it helps to facilitate collaboration in security and flexibility.

Firstly, by fixing the vulnerabilities exposed by zero-day attacks that have a tendency to target unpublished defects in computer hardware and software the suggested method maximizes public security. With a 93.94% detection rate of zero-days, a 99.07% overall accuracy, and an improved false positive ratio of 2.21%, it reduces the probability of financial loss, and leakage of information. For example, industries with large numbers of connected IoT devices, including healthcare, finance, and transportation, would obtain huge benefits. By protecting against these attacks, private information, including financial and health information, would be protected.

In addition, the flexibleness of the reinforcement learning module guarantees long-term confidence between individuals in web environments. There is a general lack of confidence in the security of the internet due to the fact that traditional signature-based defenses are invariably ineffective against constantly changing attacks. The hybrid system allows for an active defense mechanism with the cooperation of dynamic learning combined with anomaly detection. By offering scalable and cost-effective solutions that counter new threats without constant human involvement, it can empower suppressed groups such as small business or poorer countries that do not have sufficient cybersecurity resources with enhanced better cyber defense, the cyber divides could be closed and more individuals will join the global economy, filling financial gaps opened up by cybercrime. This study shows the importance of AI in creating an ethical and more inclusive society. Our framework can enable innovations such as smart cities, telemedicine, and online education to help the economy grow and protect privacy and safety because our framework can be used to secure IoT networks and ensure models are generalized across various data sources. These technologies should be affordable to everyone in order to avoid further inequality. In a summary, this thesis transcends technical development of zero day defense. It prepares a safer, more fair digital world-safeguarding humans, enhancing, and restoring confidence in our more connected systems.

3.3 Environmental Impact

We observe that anomaly detection based on AI is potentially resource-intensive and can have a high environmental footprint because of high computational costs. Our solution is specifically aimed at sustainability by optimizing the efficiency of the model (lightweight algorithms), and adaptive strategies of processing to reduce the amount of the energy consumed without the negative effect on the security performance. Such unsupervised learning algorithms as PCA and autoencoders and reinforcement learning agents, batch processing, and dimensionality reduction

algorithms are applied in a manner that they consume a minimum of memory and redundant computation. Since datasets can be enormous, and in particular, they can be in the millions, a chunk-based processing pipeline is used, which will not only make entire systems available to systems with limited resources (such as 8GB RAM) but also reduce its power consumption, and also be less harmful to the environment. The models are also targeted at the capability to be trained and run on the standard computing hardware to prevent taking the big data centers to run their work, and in the process, reduce the carbon footprint of cloud-based computing infrastructures. The incremental techniques of learning introduced into the models not only lengthen the usefulness of the models, but also decrease the level of energy needed in the long run, as well as leading to the overall sustainability of the system. This incremental addition of learning processes does not only enhance the effective life of the models, but also reduces the energy requirements over the long run, which helps in the sustainability of the system as a whole. Such widely-used endeavors form a significant component of the development of AI-powered cybersecurity solutions that will not only benefit but will not damage the environment.

3.4 Ethical Issues

Compliance with Standards and Codes of Practice

Ethical and liability issues are also critical concerns that need to be discussed to implement the proposed IDS; otherwise, it will be irresponsible. Our design complies with GDPR standards and is based on ISO/IEC 27001 information-security principles and introduces data protection and risk-management practices throughout in order to allow privacy and security. For training the models, public benchmark datasets were used (CIC-IoT2023 / CIC-BCC-NRC) instead of real time network traffic or user data. Such datasets made solely for research purposes are already anonymized by the dataset publishers ahead of release. Meaning no PII (Personally Identifiable Information) is present in the dataset and the dataset has already been ethically cleared and can in no way be traced back to any particular individual, institution, device or source; placing us on safe ground of use. Generally, CIC network traffic datasets are known to swap real IP/MAC addresses, even usernames with random tokens. The dataset publishers (University of New Brunswick — UNB) for public datasets are the data owners who in this case will ensure compliance before releasing a dataset. Regardless, we have verified that neither datasets contain any PII. No unnecessary metadata/raw packets were used in training. We have used only relevant flow-based features and dropped unnecessary columns during preprocessing. In terms of professional responsibility, the system complies with the IEEE Code of Ethics in terms of integrity, fairness, and well-being of people. The current design does not implement ISO/IEC 27001 checks or have alert formatting mechanism like using Intrusion Detection Message Exchange Format IDMEF and such; but we plan on improving the architecture so as to support standard output formats as well as integrating encryption system, access control, multi-factor authentication to align with ISO 27001. These are possible future working scopes for further ensuring compliance with standard code of practice.

3.5 Project Management Plan

This thesis was completed in a systematic way in which particular tasks have been identified, divided into overlapping phases to accommodate iterative research. Several changes to the initial idea of RL were made maintaining compatibility with available resources.

Project Schedule The Schedule for our overall project is shown in figure 3.1 as Gantt Chart.

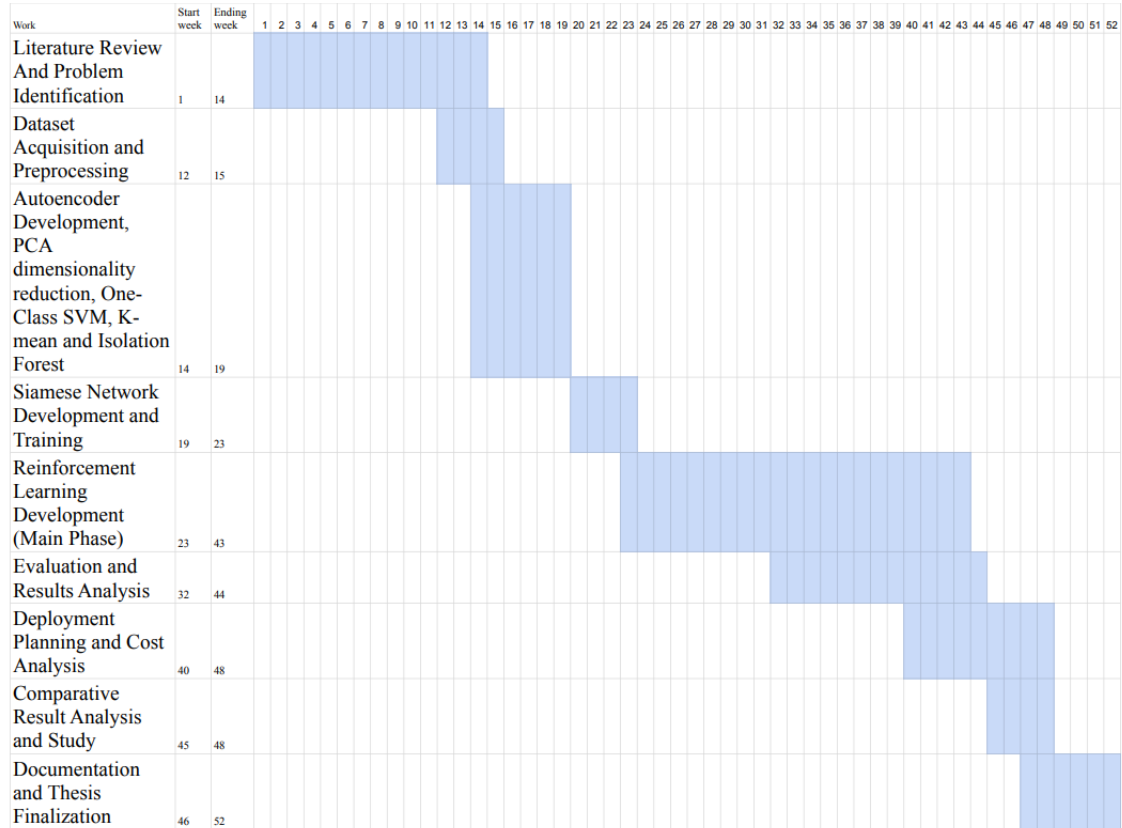


Figure 3.1: Gantt chart

Budget and Resources

Budget Consideration: Direct costs of the research phase were insignificant since all the datasets and libraries used are open-source and the computational loads are met by the usual academic resources.

Resource Allocation:

1. Technical Resources: Standard workstation with 8GB RAM and GPU.
2. Data Resources: Public benchmark datasets (CIC-IoT-2023, CIC-BCCC-NRC-2024).

3.6 Risk Management

There are a number of risks that have been identified and mitigation plans to be put in place that will see the research successfully completed. Data imbalance, where zero-day samples are often underrepresented in the data, is one of the biggest risks; it can be mitigated by using anomaly detection methods and Siamese neural networks, which are trained to make predictions outside the restricted set of labeled samples. A second risk is overfitting of the model, where the system may pick up noise particular to the training data. To overcome this, regularization methods such as dropout and cross-validation, as well as validation on two divergent datasets, are employed in order to reduce overfitting. A concern on computational matters also exists since the size of the training data is huge and it may exceed the memory capabilities. This is addressed by the use of chunk-based training and dimensionality reduction software to reduce the memory footprint. In addition to reducing the risk of extremely high false positives that can overwhelm the user, it can choose reinforcement learning reward schemes to discourage false alarms and promote moderate detection performance. The complexity of the algorithm development and debugging enhances the probability of delays in implementation; to address this, the project timeline is integrated with standalone development phases, consistent development verification, and buffer time parameters. At last, there is the possibility that the suggested hybrid AI system could fall short of the proposed 99% accuracy in zero-day attacks. To this end, several algorithmic methods are studied such as PPO, DQN and SAC, with extensive hyperparameter optimization and cross-validation steps. The performance fallback level has been set to 96% accuracy, which is considered to allow acceptable detection as the system remains effective in the event that the primary target is not achieved.

3.7 Economic Analysis

The proposed Intrusion Detection System (IDS) is highly cost-effective in the development and implementation process. The datasets were all open-source at the time of conducting the research, i.e., CICIOT2023 dataset and CIC - BCCC - NRC TabularIoTAttacks 2024 dataset, and the entire system had been created in open-source, i.e., TensorFlow, PyTorch, and Scikit-learn, without any licensing costs. These tests were conducted in one of the regular workstations (Intel i7, 8 GB RAM, GTX 1650 GPU) with additional free access to the cloud GPU environments. In order to do so in practice, intermediate costs are acquired on the grounds of calculation facilities, data storage, monitors, and model maintenance. However, it is highly cost-efficient, as the model is estimated to cost USD 4 500-7 000 per year, which is a lot less than the commercial IDS systems, which are usually in the USD 20 000 range or more. The projections of the estimated spending on a yearly basis are as shown in Table 3.8. The system is economically valuable since it has a low development cost and scalable operation cost that reduces the downtime, autonomous adaptability, and long-term cost savings against the losses that might be incurred as a result of zero-day attacks.

Cost-Benefit Estimation

Costs: The major costs are moderate costs to infrastructural costs, computing, and storage.

Benefits:

1. The system is likely to reduce massive expenditures against zero-day attacks that may cost millions of dollars in various industries.
2. It lowers the downtimes of IoT infrastructures, which increases the business continuity and service availability.
3. It provides adaptive learning features through reinforcement learning, which means that people do not need to update the rules often.

Return on Investment (ROI) Because of the low outlays of its development and medium cost of deployment, the IDS provides high economic results by:

1. Avoiding financial losses that are related to security breaches.
2. Ensuring advanced operational efficiency and protection with the help of automated and flexible defenses.
3. Tying the rope around operational spending in the long term through incremental educational learning as opposed to wholesale retraining.

Overall, the given solution is economically viable both in the research phase and in the further practical implementations. In Table 3.1, the estimated deployment cost of the proposed IDS is shown below.

Table 3.1: Estimated deployment cost of the proposed IDS

Cost Component	Description	Estimated Annual Cost (USD)
Compute Infrastructure	Servers or cloud GPU instances for real-time traffic analysis	\$2,000 – \$5,000
Data Storage & Backup	Storage of logs, anomalies, and retraining datasets	\$500
Integration & Monitoring	Security analytics and system integration tools	\$1,000
Maintenance & Updates	Model retraining, anomaly threshold tuning, security patches	\$1,000
Total Estimated Cost	—	\$4,500 – \$7,000

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

The overall objective of this work is to design and build a hybrid, dynamic, and intelligent intrusion detection system, which is capable of being aware of unknown threats as well as normal threats i.e. zero-day attacks. To achieve this, the design process is organized into multiple consecutive stages to successfully screen out unseen anomalies.

1) **Model Training by Dataset:** We will be training two models with two different datasets which will later be used in the Siamese network.

- **Model Training using CICBCCNRCTabularIoTAttacks2024:** The approach began with Principal Component Analysis (PCA) on the CICBCCNRCTabularIoTAttacks2024. The technique was selected after trying multiple other machine learning algorithms, and after evaluating performance PCA was shown to perform the best on this particular dataset. It projects a high-dimensional dataset onto a set of novel, independent features known as principal components, retaining useful information. This is particularly useful for anomaly detection.
- **Model Training using CIC IoT Dataset 2023 dataset:** For this particular dataset we chose Standard autoencoder after evaluating performance metrics of several different models including PCA-Based Anomaly Detection, Isolation Forest, One-Class SVM and K-mean and finding Standard autoencoder to be the best fit. Standard autoencoders have the ability to reconstruct normal data patterns and significant deviations from learned patterns give it a particular benefit in successfully detecting anomalies.

2) **Siamese Network:** A siamese network is used to compare inputs and detect similarities between twin networks - two identical neural networks. These twin

networks learn by comparing instance pairs from each input branch and adjusting weights to either minimize or maximize distance between similar pairs or use some other calculation metric for adjusting the neural network.

3) **Reinforcement Learning:** Training of Proximal Policy Optimization (PPO) agent is performed with output of the Siamese network, and it learns a detection choice adaptively. The reward function pilots balanced performance by assigning reward +1 for correct identification of rare unseen attacks and penalty -1 for false negatives.

4) **Testing:** Attack classes are withheld from both datasets to be used exclusively for testing to evaluate generalization and unseen attack detection capability. General performance metrics; accuracy, precision, recall, F1-score, ROC, and Zero-Day Detection Rate (ZDR) as well as RL specific parameters like discount factor, total reward, clipping epsilon and convergence rate are automatically generated. The results demonstrate model adaptability.

In short, the approach is crafted for detection, learning and adaptability.

4.2 Preliminary Design or Design (Model) Specification

This study compared performance of five unsupervised machine learning models in identifying anomalous and the zero day attacks on an IoT traffic network. These models are:

- PCA-Based Anomaly Detection
- Autoencoder
- Isolation Forest
- One-Class SVM
- K-means Clustering

The analysis is performed on two popular IoT datasets, namely the CIC IoT Dataset 2023 and the CICBCCNRCTabularIoTAttacks2024. The two datasets were both preprocessed properly to be used to model unsupervised anomaly detection. Because the models are used mainly in a semi-supervised or unsupervised mode, only the benign (normal) traffic samples were trained. The methodology also permits the models to adopt typical normal network behaviors hence enabling them to detect abnormalities that point to anomalies or attacks.

In order to have the best evaluation, every dataset was separated into two distinct subsets training and testing. Between 1,000,000 benign samples each were used in

the training sets, and the testing sets contained about 4,000,000 samples, a combination of benign and labeled attack traffic.

The CIC IoT 2023 dataset includes approximately 40 well-chosen flow and packet features, including but not limited to the flow duration, TCP and UDP flags, protocol type, packet rates, and statistical moments of traffic metrics (mean, standard deviation, minimum, maximum). The Attack type distribution of CIC-IOT2023 is shown in figure 4.1

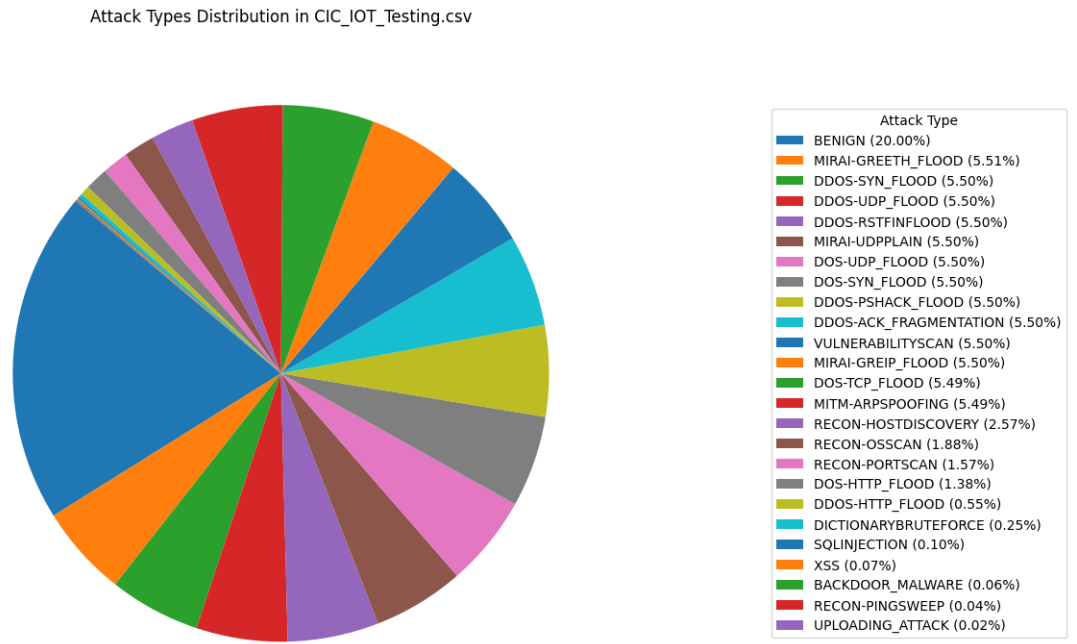


Figure 4.1: Attack types distribution in CIC-IOT23

Conversely, CIC-BCCC-NRC-TabularIoTAttacks-2024 is more detailed, sparse in its feature space of more than 80+ features of traffic. Such features are detailed flow metrics, packet header statistics, timing information, and fine-grained flag counts, as obtained with advanced network flow analyzers such as CICFlowMeter. The Attack types distribution in CIC-BCCC-NRC-TabularIoTAttacks-2024 is shown in figure 4.2

Attack Types Distribution in CIC-BCCC-NRC-Testing.csv

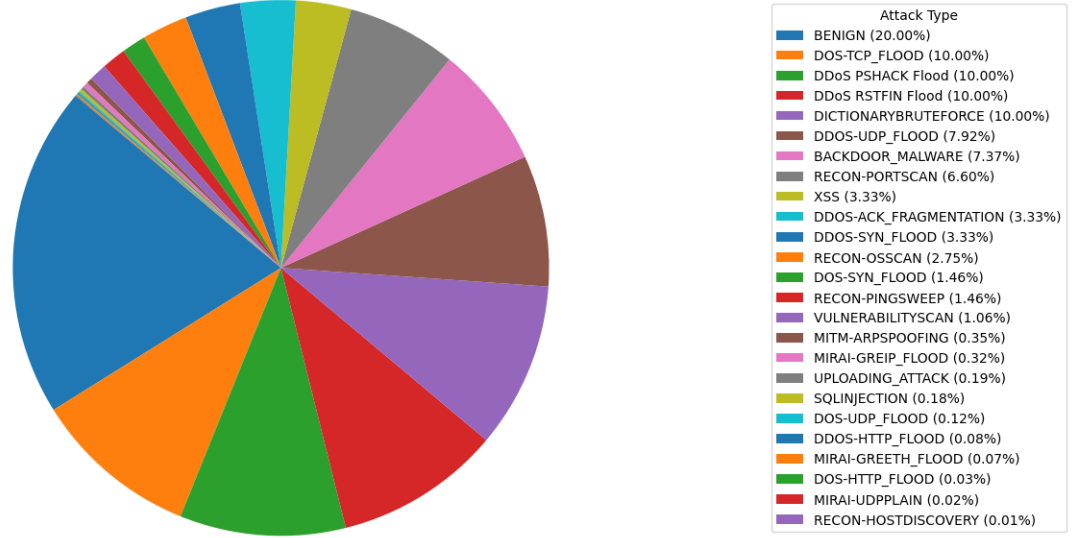


Figure 4.2: Attack types distribution in CIC-BCCC-NRC-TabularIoTAttacks-2024

After thorough preprocessing and harmonization the features recognized that were similar or semantically similar were approximately 20 between the two datasets. This subset of common features enabled the comparison in cross-dataset and modeling transferability.

The testing was aimed at comparing these five unsupervised models on their performance in detecting anomalies in the two datasets, gauging their capacity of generalizing to different forms of attacks, as well as, reducing the false positives. In conclusion, our final goal is to discover and integrate the best available models into a hybrid and dynamic intrusion detection system for real-time zero-day attack discovery for the IoT domain.

CIC IOT Dataset2023:

Performance Evaluation of the Isolation forest,PCA,Kmean and One class SVM model:

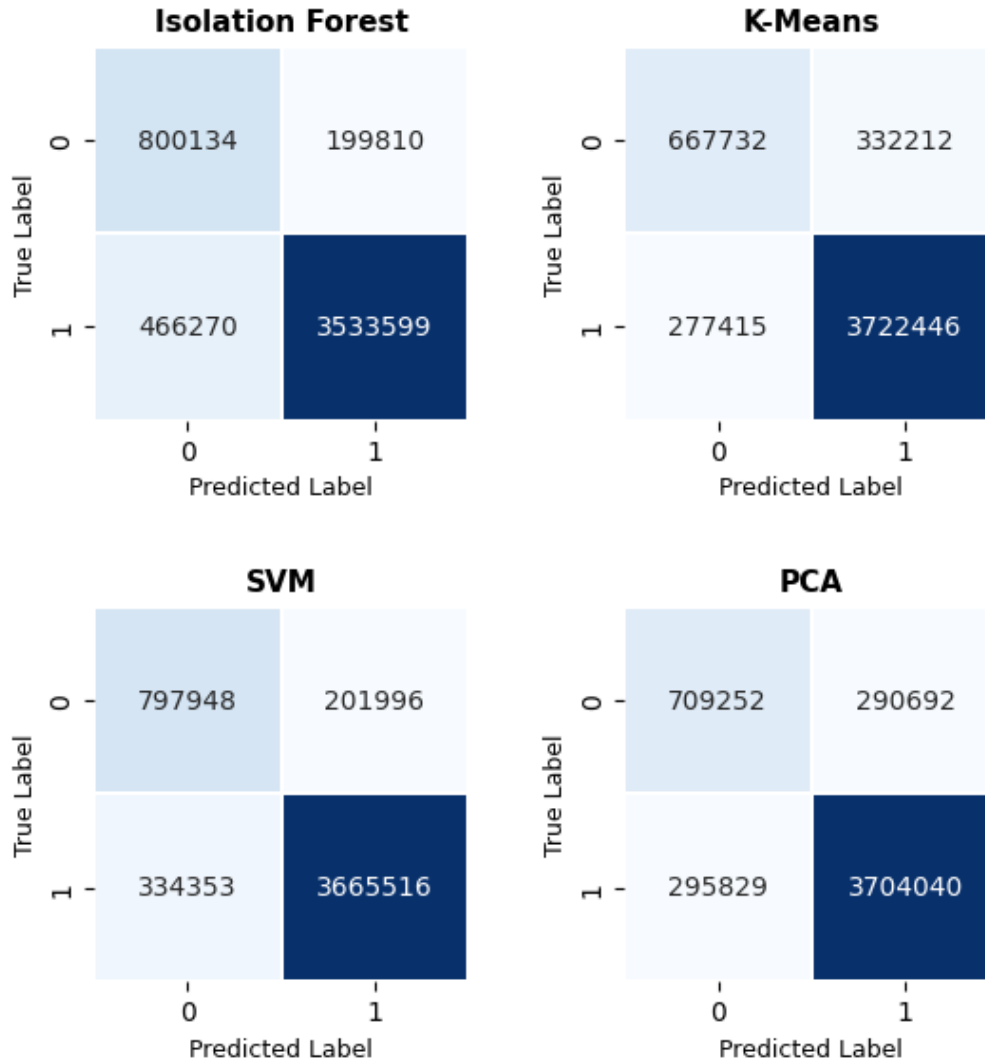


Figure 4.3: Confusion matrices of Isolation forest,SVM,Kmean and PCA of CIC IOT 2023

In figure 4.3, the confusion matrices of Isolation forest is shown. Here,true Negatives (800134 correct Benign),False Positives (199810 misclassified Benign),False Negatives (466270 missed Attacks) and True Positives (3533599 correct Attacks). Total: 499813 samples. Accuracy=88%. For Kmean,true Negatives (667732 correct Benign),False Positives (332212 misclassified Benign),False Negatives (277415 missed Attacks) and True Positives (3722446 correct Attacks). Total: 4999805 samples. Accuracy=88%. For One class SVM,true Negatives (797948 correct Benign),False Positives (201996 misclassified Benign),False Negatives (334353 missed Attacks) and True Positives (3665516 correct Attacks). Total: 499813 samples. Accuracy=89%, For PCA,true Negatives (709252 correct Benign),False Positives (290692 misclassified Benign),False Negatives (295829 missed Attacks) and True Positives (3704040 correct Attacks). Total: 4999813 samples. Accuracy=88%.

Table 4.1: Evaluation metrics of all the five unsupervised models of CIC-IOT-2023

Algorithm	Accuracy	Precision	Recall	F1-score	Anomaly %	Runtime (min)
Autoencoder	0.89	0.88	0.89	0.88	82.78%	7.26
Isolation Forest	0.88	0.88	0.88	0.88	80.89%	4.86
K-Means	0.88	0.88	0.88	0.88	81.10%	4.55
One-Class SVM	0.89	0.87	0.89	0.88	81.03%	20.12
PCA	0.88	0.87	0.88	0.87	83.03%	4.15

In the case of CICIOTDataset2023, Our priority was the overall F1/accuracy and attack detection power in particular if we look at Table 4.1. The Standard Autoencoder algorithm performs best here as it gave:

- Highest accuracy (0.89, tied with One-Class SVM, slightly better than others),
- Highest F1 for attack,
- Highest recall for attack.
- Fast training compared to SVM, though slightly slower than IF/K-Means/PCA

Selected Model (Autoencoder):

Performance Evaluation of the selected model:

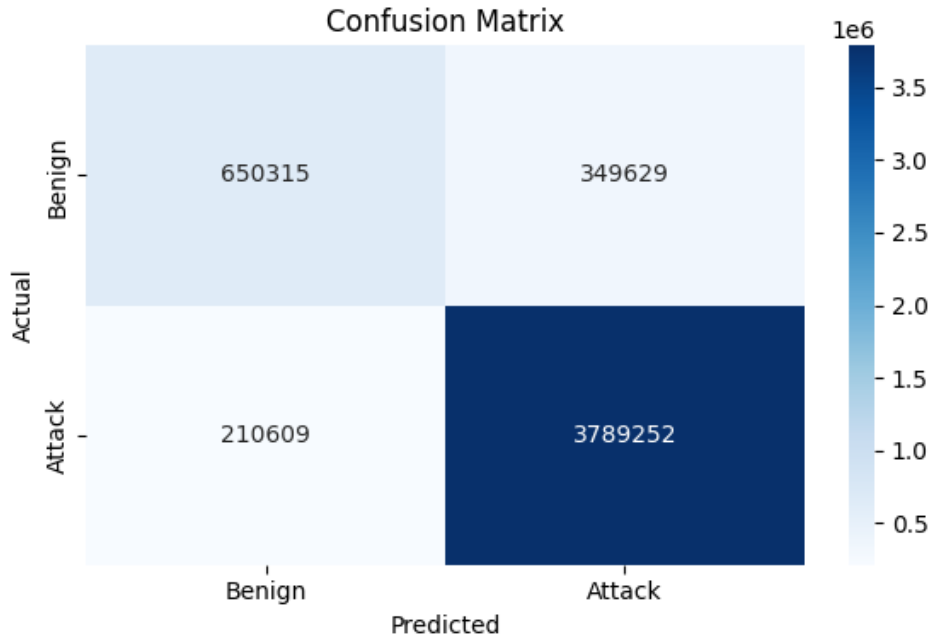


Figure 4.4: Confusion matrix (Standard Autoencoder)

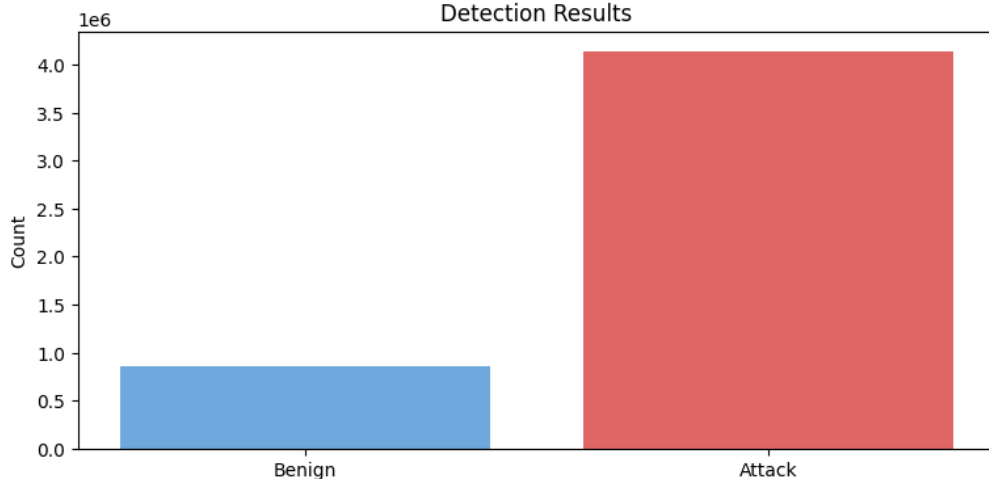


Figure 4.5: Detection Results of Autoencoder CIC IOT 2023

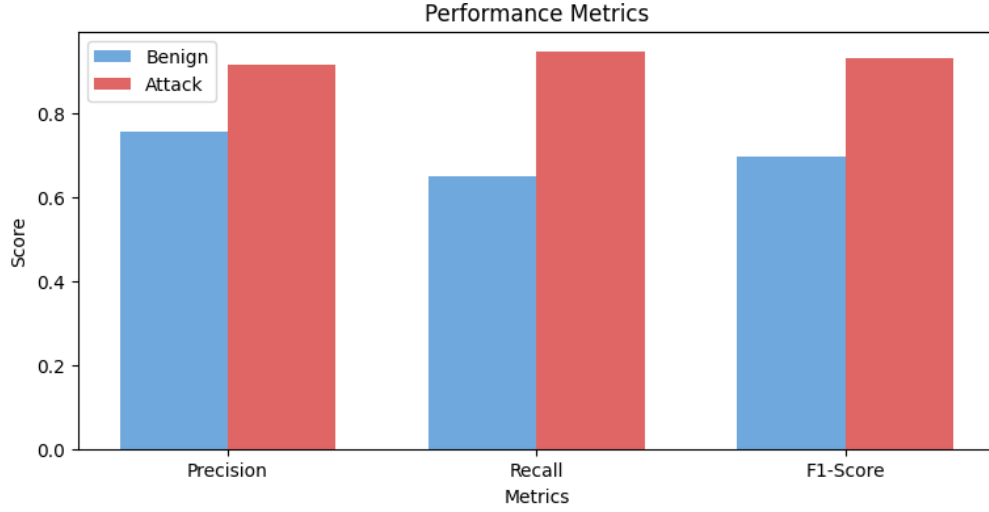


Figure 4.6: Performance metrics of autoencoder CIC IOT 2023

In figure 4.4, the confusion matrix of standard autoencoder is shown. Here, true Negatives (650315 correct Benign), False Positives (349629 misclassified Benign), False Negatives (210609 missed Attacks) and True Positives (3789252 correct Attacks). Total: 4999805 samples. Accuracy=89%

In figure 4.5 and 4.6, the amount of benign is 1 million and the amount of attack is 4 million. The precision, recall and F1 of benign traffic are 0.76, 0.67, 0.71 respectively. While Attacks excel with high precision, recall and F1 which are 0.92, 0.95, 0.93 respectively.

CIC-BCC-NRC-TabularIoTAttacks-2024:

Performance Evaluation of the Isolation forest,Standard Autoencoder,Kmean and One class SVM model:

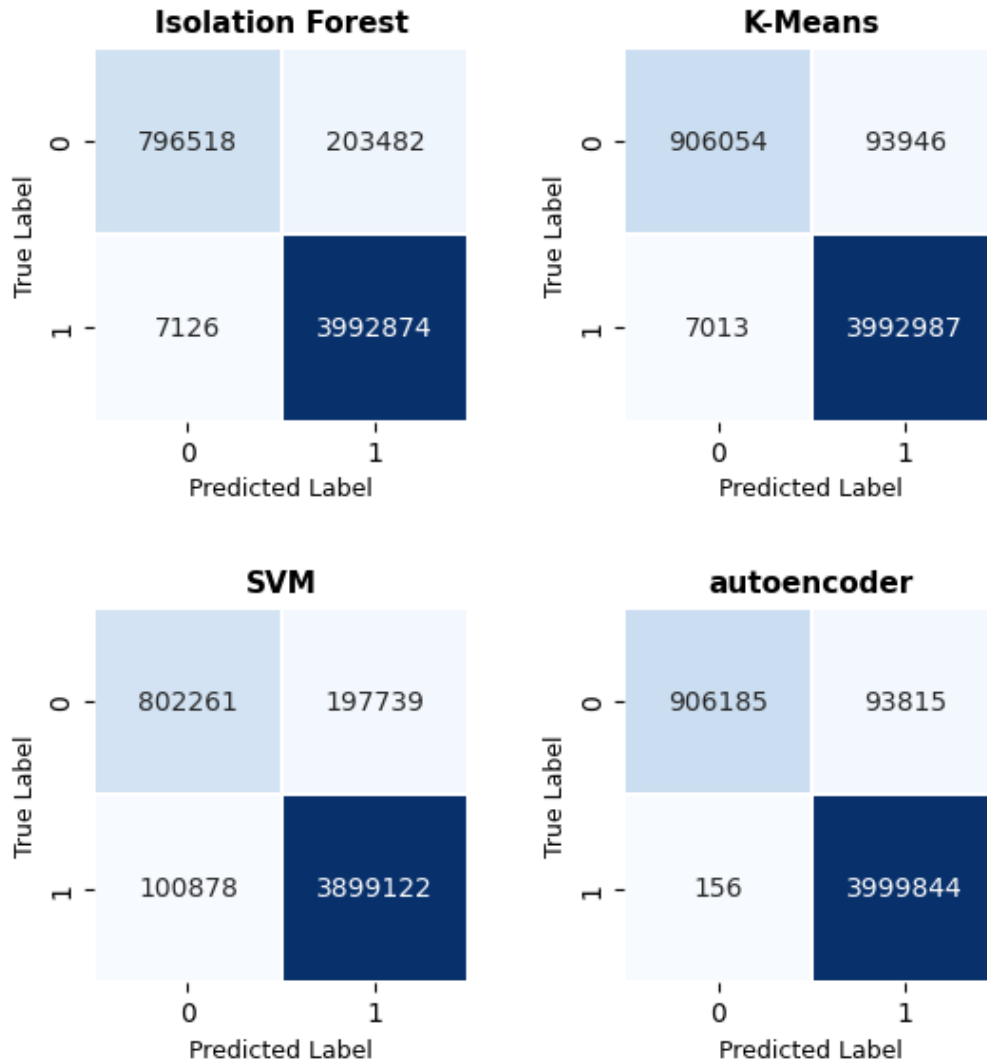


Figure 4.7: Confusion matrices of Isolation forest,SVM,Kmean and autoencoder of CIC-BCC-NRC-TabularIoTAttacks-2024

In figure 4.7,the confusion matrices of the standard autoencoder is shown.Here,true Negatives (906185 correct Benign),False Positives (93815 misclassified Benign),False Negatives (156 missed Attacks) and True Positives (3999844 correct Attacks). Total: 500,000 samples. Accuracy=97% . Now, for Isolation forest,true Negatives (796518 correct Benign),False Positives (203482 misclassified Benign),False Negatives (7126 missed Attacks) and True Positives (3992874 correct Attacks). Total: 500,000 samples. Accuracy=95% .For KMean,true Negatives (906054 correct Benign),False Positives (93946 misclassified Benign),False Negatives (7013 missed Attacks) and True Positives (3992987 correct Attacks). Total: 500,000 samples. Accuracy=97% .For One class SVM,true Negatives (802261 correct Benign),False Positives (197739 misclassified Benign),False Negatives (100878 missed Attacks) and True Positives (3899122 correct Attacks). Total: 500,000 samples. Accuracy=94% .

For CIC-BCCC-NRC-TabularIoTAttacks-2024, Autoencoder, K-Means, and PCA all produce nearly identical, top-of-the-class results as shown in Table 4.2.

- Accuracy, Precision, Recall, F1-score: 0.98
- Very balanced benign/attack detection.
- Autoencoder is slightly low by anomaly detection % and runtime.
- PCA works best in this case of all the models.

Table 4.2: Evaluation metrics of the five unsupervised anomaly detection algorithms of CIC BCCC NRC TabularIoTAttacks 2024

Algorithm	Accuracy	Precision	Recall	F1-score	Anomaly %	Runtime (minutes)
PCA	0.98	0.98	0.98	0.98	81.87%	6.84
Autoencoder	0.97	0.98	0.98	0.98	81.83%	9.14
Isolation Forest	0.95	0.95	0.95	0.95	84.35%	7.67
K-Means	0.97	0.98	0.98	0.98	81.74%	6.92
One-Class SVM	0.94	0.95	0.97	0.96	81.03%	33.18

Selected Model (PCA):

Performance Evaluation of the selected model:

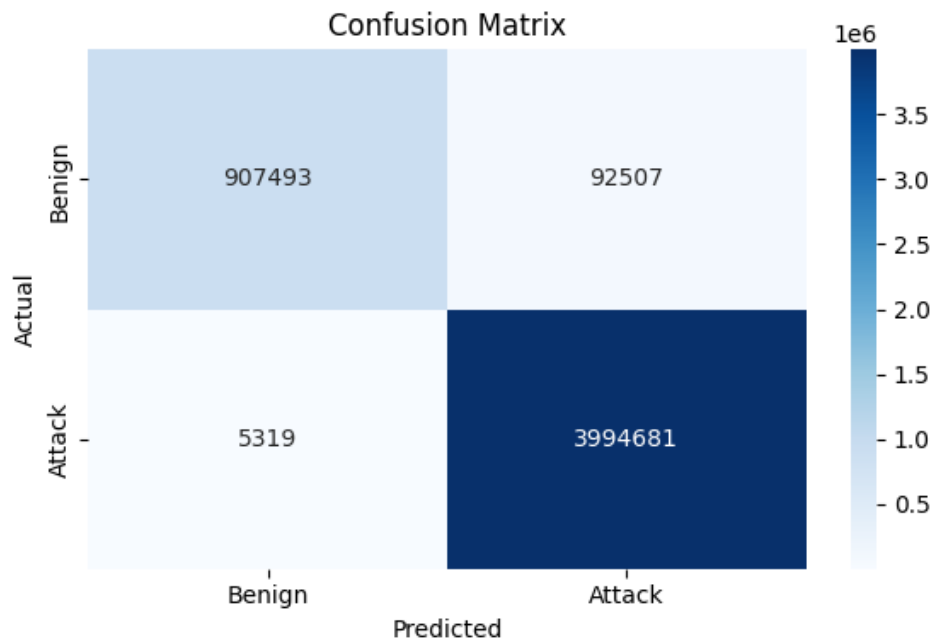


Figure 4.8: Confusion Matrix (PCA)

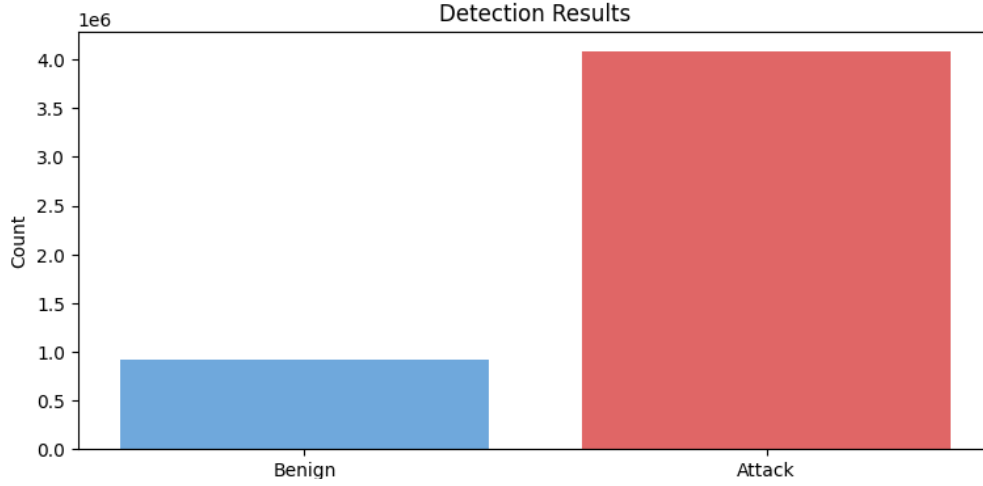


Figure 4.9: Detection Results of PCA of CIC-BCCC-NRC-TabularIoTAttacks-2024

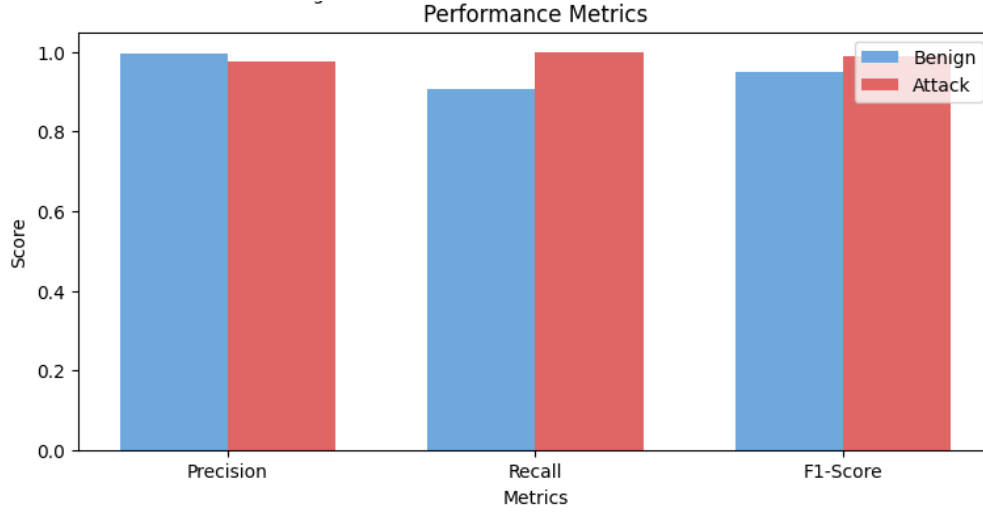


Figure 4.10: Performance Metrics of PCA of CIC-BCCC-NRC-TabularIoTAttacks-2024

In figure 4.8, the confusion matrix of PCA is shown. Here, true Negatives (907493 correct Benign), False Positives (92507 misclassified Benign), False Negatives (5319 missed Attacks) and True Positives (3994681 correct Attacks). Total: 500,000 samples. Accuracy=98%

In figure 4.9 and 4.10 the amount of benign is 1 million and the amount of attack is 4 million. And the Benign traffic shows near perfect Precision which is 1.00, recall, F1 are 0.91, 0.95, while attacks excel with perfect recall which is 1.00, precision and F1 are 0.98, 0.99 respectively.

4.3 Data Collection

For this paper we use two different IOT datasets, CIC-BCCC NRC TabularIoT 2024 and CIC IoT 2023; both from the official Canadian Institute for Cybersecurity (CIC) repository, publicly accessible to everyone for academic and research purposes.

CIC-BCCC-NRC-TabularIoTAttacks-2024 dataset: This is an augmented collection, combining nine common IoT datasets. It is labeled and tabular containing over 1 million benign and malicious network traffic data collected from IoT devices and includes detailed feature extraction using CICFlowMeter.

CIC IOT 23 dataset: A comprehensive collection of 11 different types of DDOS attack, 1 brute force attack, 2 kinds of spoofing attack, 4 kinds of DOS attack, 5 kinds of Recon attack, 6 kinds of Web-based attack and 3 kinds of Mirai attack are captured in individual CSV files with detailed network attributes including flow statistics, packet details, TCP flags and protocol behaviors. This dataset was selected due to its diversity of realistic attacks, extremely relevant with current IoT context and well balanced benign and malicious data. Its availability and the popularity in recent academic studies makes it a reproducible and believable benchmark.

Elaboration included renaming columns to be consistent, combining similar features as needed and ensuring categorical variables, like the protocol types, were aligned across the datasets. This matching enabled us to develop a consistent feature space of both datasets, which is necessary to support cross-dataset comparison as well as model training. Additionally, 20 attacks were common in the two datasets, despite the fact that CIC IoT 2023 dataset included 32 attack types, and the CIC-BCCC NRC 2024 dataset included 28 attack types. Using these, we derived parity testing sets, to conduct performance and generalizability assessment of selected models consistently on the two datasets.

4.3.1 Data Cleaning

We used several techniques for data cleaning to ensure that the datasets were consistent and suitable for unsupervised anomaly detection. First of all, machine learning algorithms can't process infinitive values which appear during feature extraction or computation. We replaced these infinite values with NaN so that they can be handled uniformly. Machine learning algorithms also can't handle missing values (NaN) if they are not specifically supported. We removed those rows containing missing values. We removed this after replacing Inf. This ensures that the training dataset contains valid and complete only. This also reduces our dataset size. In both the training and testing datasets, the label column was retained for evaluation purposes; however, it was dropped at the time of applying the unsupervised algorithms so that the models were trained without label information. This is not a traditional data cleaning step but it is important to our quality control. We also

used z-scores to find and select values that are significant. We also Identified and removed duplicate rows. As we made the training dataset as balanced as possible, no additional balancing techniques were required in the current work. However, for larger datasets in future research, more advanced data cleaning methods may be applied. These include Winsorization, which caps extreme values at specific percentiles instead of removing them; Feature Variance. Thresholding, which removes features with SMOTE (Synthetic Minority Oversampling Technique), near-zero variance. Imbalance in classes are addressed by artificially generating minor class entries; and Anomaly Score–Based Cleaning, which assigns anomaly scores (e.g., using Isolation Forest) to the training data and removes only those points with exceptionally high scores.

For reinforcement learning, all collected transitions were first screened to replace any infinite values with NaN and drop those entries so the replay buffer contains only valid samples; state features and logged metrics were normalized online using the same StandardScaler applied in pretraining; rewards were clipped to a finite range to prevent instability, and latency and resource-usage values were similarly bounded; during mini-batch sampling, transitions with unusually high TD-errors (beyond three standard deviations) were filtered out to avoid corrupt updates; finally, advantages and returns were standardized to unit variance with mean zero before each policy update to ensure stable and reliable learning dynamics.

4.3.2 Data Transformation

We also applied feature scaling using a standard scaler. Features underwent normalization by deducting the mean before division with standard deviation, which produced zero-centered values with a unit variance. This step was crucial to ensure that features with large magnitudes, such as Tot sum, did not dominate smaller-scale attributes. Standardization allows equal contribution of all features to the learning process. This is particularly important for reconstruction-based and distance-based unsupervised models. In addition, feature selection was performed manually, keeping meaningful information only. This in turn reduces dimensionality and improves computational efficiency. Selective retention of features significantly improves anomaly detection by minimizing noise and overall redundancy.

The data further goes through some degree of transformation before it is received by the RL module. First, a particular set of features relevant to the RL task is selected while unnecessary columns like "Flow ID," "Src IP," "Dst IP," and "Timestamp" are dropped off. Categorically labeled columns are transformed to numerically encoded labels appropriate for the model's interpretation. 'Attack' traffic is marked 1 and 'Benign' traffic 0. Numerical features are then scaled to ensure equal contribution of all features in the learning process. Finally, StandardScaler is used for feature normalization for standard deviation of 1 and mean 0.

4.3.3 Data Integration

Data integration involved cross learning from two different IoT datasets; CIC IoT Dataset 2023 and CICBCCCNRCTabularIoTAttacks2024. Each dataset was designed for different applications but both originating from IoT network environments barely differed in form or naming convention. CIC IoT Dataset 2023 dataset is a widely used benchmark dataset with benign and malicious IoT traffic that is labeled, while the CICBCCCNRCTabularIoTAttacks2024 dataset is a recent one that provides recent IoT traffic patterns and more accurate attack classes. The combination of them provided a more plentiful feature space and more varied attacks, which enhanced the credibility of anomaly detection experiments. The process of integration began with the identification of a common feature space across the two datasets. Through schema matching and feature alignment, it was observed that several flow-based attributes were shared, including Header_Length, Protocol Type, Rate, Tot sum, Min, Max, AVG, Std, Variance, Number, IAT, and multiple TCP flag indicators (fin_flag_number, syn_flag_number, rst_flag_number, psh_flag_number, ack_flag_number, ece_flag_number, cwr_flag_number). The class label field (Label) was also present in both datasets, but it was dropped in the model training phase because model training of this dataset was done by unsupervised models. Certain fields required normalization for compatibility. For example, the Protocol Type attribute used different encodings in the datasets, which were harmonized to a numeric convention (ICMP = 1, TCP = 6, UDP = 17) to guarantee consistency. Integration was performed cautiously by making sure that both datasets had passed through the same preprocessing steps like removal of missing values, enforcement of numeric types, and Min-Max scaling so that the resulting final combined feature space can be easily comparable in subsequent steps, especially while distancebased anomaly matching. This integration was essential for enabling robust anomaly detection, particularly in crossdataset and zeroday attack detection scenarios, where consistency and comparability across data sources are crucial.

The data integration process begins by loading the training dataset (RL Attack Train.csv) with 17 feature columns—such as Header_Length, Rate, various TCP flag numbers, and statistical metrics like Tot sum, Min, Max, AVG, Std, Number, Variance, and IAT—and the unseen attack dataset (Unseen_Attack_Dataset_22801.csv) containing 19 columns, including extraneous 'Tot size' and 'Label' fields. To ensure compatibility, the unseen dataset's labels (19 distinct attack types, evenly distributed at 1,200 samples each) and 'Tot size' are promptly removed, aligning it to the exact 17-column structure of the training data. With a prefitted scaler from training, selective normalisation is subsequently applied for four significant columns (Rate, Tot sum, AVG, and Std). This preserves the binary 0–6 range of flag columns like fin_flag_number while moving values to match the training distribution (e.g., means near 0 and std near 1). We created a clean and normalized Unseen Attack Dataset with 22,800 samples of IoT traffic, which is stored as Unseen Attack Dataset Normalized.csv. This CSV file can be directly injected into the reinforcement-learning environment without additional processing.

4.3.4 Data Reduction

We used manual feature selection to decrease the number of the features dimensionality to facilitate analysis and decrease the time required to process the features, and also to ensure that the feature selection was consistent across the datasets. The most informative attributes were then only retained and reduced the amount of computation thus important signal information was not lost. Domain knowledge and reasoning were the premise for feature selection. Certain TCP flag indicators (fin_flag_number, syn_flag_number, rst_flag_number, psh_flag_number, ack_flag_number, ece_flag_number, cwr_flag_number) of both data sets have been maintained with Header_Length, Protocol Type, Rate, Tot sum, Min, Max, AVG, Std, Variance, Number, and IAT. Complete anomaly detection is made done by the combination of these features, which capture a balanced blend of traffic strength, statistical distribution, periodic pattern, protocol behavior, and TCP control signal usage. Mixed traffic (benign + attack) was held in reserve for test, while benign traffic alone was held over for training the unsupervised models to learn an initial impression of normal behavior. Similarly, few of the model anomalies that had shown significant discrepancy with the second set of data which is the CIC BCCC NRC tabularIoT dataset 2024 were forwarded for investigation in the final distancebased filtering (Euclidean distance thresholds). Three distance based metrics were done in the final distancebased filtering which are cosine, manhattan and euclidian and from these euclidian was selected because of the good performance. Distancebased filtering was a secondlevel reduction mechanism that focused on selecting out unusual anomalies, which are of greatest interest in zeroday detection scenarios. Thus, the combined strategy of selective feature retention, training on normal data, and distancebased anomaly filtering formed a unified data reduction pipeline that was both computationally efficient and analytically focused.

4.3.5 Summary of Preprocessed Data

Following extensive preprocessing and integration, two preprocessed and fully normalized cleaned datasets were obtained: Anomalies Autoencoder 2023.csv and PCA Anomalies 2024.csv. Anomalies Autoencoder 2023.csv and PCA Anomalies 2024.csv files contain samples of network traffic that were identified as anomalies by an unsupervised algorithms. Standard autoencoder algorithm and PCA algorithm trained on the CICIOT2023 dataset and CIC - BCCC - NRC TabularIoTAttacks 2024 dataset respectively. For consistency, the characteristics of the samples that map to a combination of recognizable categories of attacks have been normalized. The normalized columns are Header_Length, Protocol Type, Rate, Tot sum, Min, Max, AVG, Std, Variance, Number, IAT, and some TCP flag predictors (fin_flag_number, syn_flag_number, rst_flag_number, psh_flag_number, ack_flag_number, ece_flag_number, and cwr_flag_number) have been normalized into a consistent format and structure. Missing and inconsistent values were addressed during this preprocessing step, all numeric attributes were Min-Max normalised, and protocol types were remapped to their respective numerical equivalents (ICMP to 1, TCP to 6, UDP to 17). To compare the distance metrics, both were preprocessed and subsequently projected onto common feature columns. The process allowed a closer study of the corre-

spondence between the model-identified anomalies and previously known malicious activity. The last anomalies, which were found by using this comparison, were the ones that were largely dissimilar to the previously observed attack patterns, which are potential zero-day threats. The Final Uncommon Anomalies.csv saves these new anomalies, and they are utilized to identify network traffic that do not comply with any of the threat profiles that are known, providing another point of anomaly filtering.

4.4 Implementation of Selected Design

The system proposed works in a multi-phase pipeline that combines unsupervised learning and dimensionality reduction and statistical anomaly filtering, as shown in the methodology diagram.

1) **Cross-Dataset Anomaly Filtering via Siamese Architecture:** The next important step after extracting anomalies on the PCA and standard autoencoder modules is to isolate them with high-fidelity, which is aided by a Siamese Network. The Siamese Network does not merely act as a cross-dataset comparator but also as a primary filtering mechanism in order to detect anomalies that are most likely to represent indeed new attack behavior. The primary goal of this step is to narrow down and reduce the outputs to a high-fidelity dataset with only the most dissimilar and rare anomalies, which are the possible manifestations of zero-day attacks. This filtering is critical, in order to train an adaptive agent, focusing its learning on novel and challenging threats rather than common ones.

Feature Normalization and Preparation: It has been observed in both of the datasets that not all the columns were normalized in those. Among all the columns some needed to be normalised such as “Header length”, “Rate”, “IAT”, and “Number”. So, these are normalized within the Siamese pipeline. This ensures metric outputs are interpretable, with every feature contributing equally to the dissimilarity calculation.

Distance Metric Optimization and Scoring:

- **Euclidean Distance:** Straight line displacement measure of two points in the feature space.
- **Manhattan Distance:** Sum of absolute differences of coordinates or distance between two points in a grid-based system, also referred to as city-block distance.
- **Cosine Distance:** Measures the cosine of the angle between two vectors, focusing on the orientation of the data points rather than their magnitude.

Each anomaly pair was scored with all three metrics, and the distance functions were evaluated for their effectiveness in identifying rare, high-risk anomalies. The evaluation metrics of the distance metrics models are shown in table 4.3.

Table 4.3: Evaluation metrics of the three distance-based similarity models.

Distance Metric	AUC	F1-Score	Spread
Cosine	0.4985	0.3320	0.0778
Manhattan	0.5046	0.3323	170022.4141
Euclidean	0.5048	0.3323	170026.6319

Every distance metric underwent a full ROC evaluation. Euclidean distance yielded the highest AUC (0.5048), F1 score (0.3323), and greatest “spread” (170,026), meaning it best maximized the separation between shared and truly unique attacks.

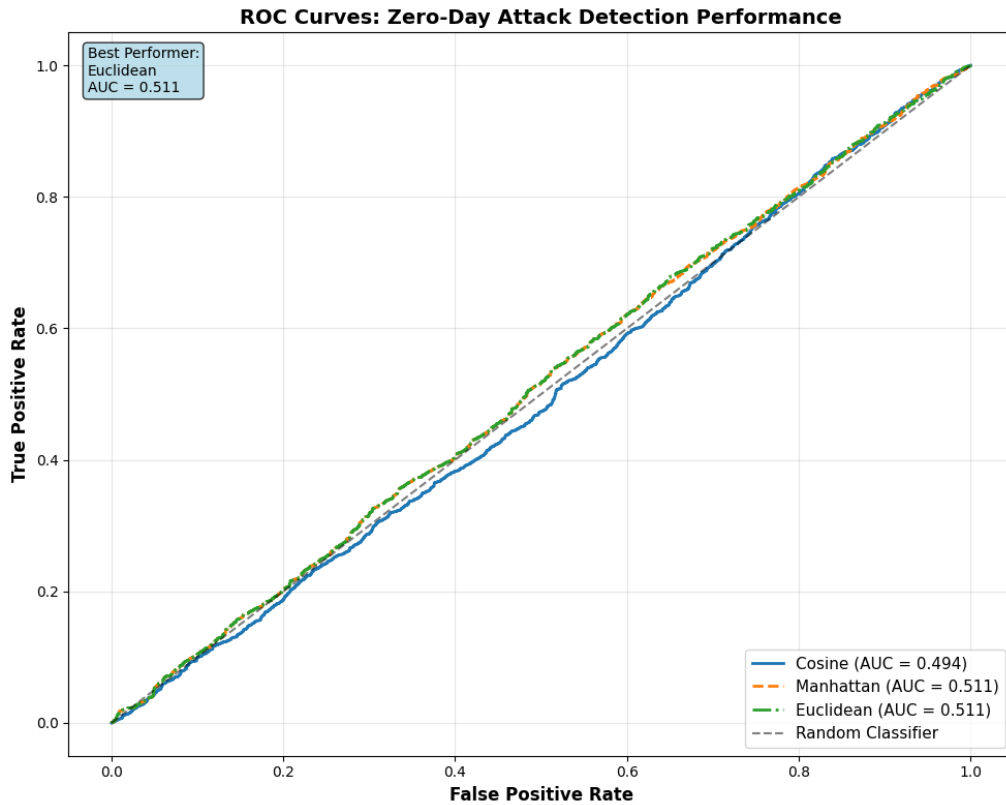


Figure 4.11: ROC curve of the three distance metrics models

In figure 4.11, the AUC score is approximately 0.5 since the dataset was very imbalanced. Attack type count was significantly higher than benign count.

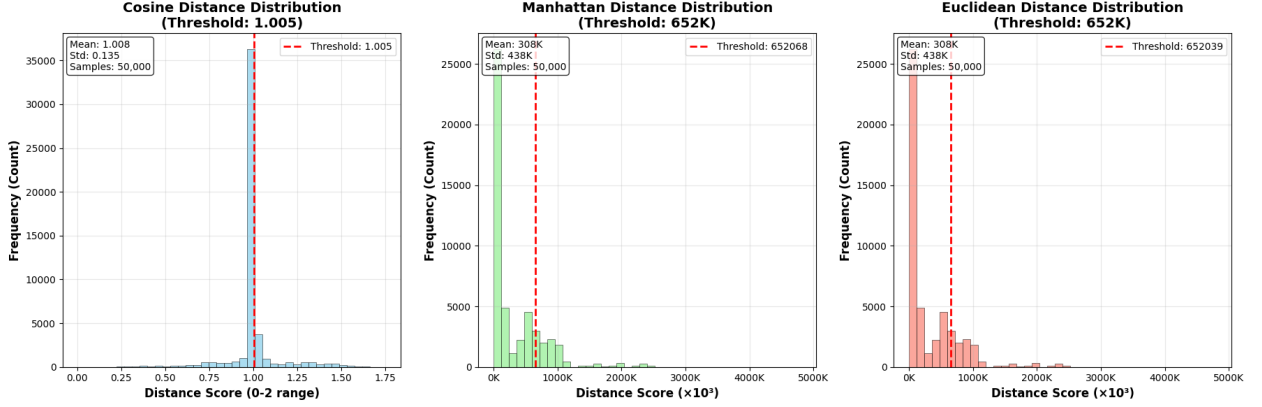


Figure 4.12: Distance Metrics Comparison

In figure 4.12, To detect the potential novel attacks, this three-panel histogram compares the Cosine, Manhattan, and Euclidean distances between scaled 2023-2024 network anomaly features (50K samples). Abrupt spike at 1.0 reflecting high direction similarity with very few departures above threshold 1.004 is evident through the cosine (mean 1.006, std 0.131). Between 0-350K, Manhattan (mean 310K, standard deviation 442K) and Euclidean (identical statistics) possess right-skewed decays with tails indicating possible anomalies at threshold values of 653K. Others emphasize magnitude spreads, while cosine excels in compactness and extracts about 10,000 rare anomalies for varied RL training (e.g., 57% DOS-TCP_FLOOD).

Final Anomaly Extraction: Applying the optimal metric and threshold produced the finalized anomaly set for downstream RL training. The deliberately skewed distribution confirms that the pipeline successfully isolated rare, novel, and high-consequence threats. These labeled outputs, by attack type and sample count, become the foundational inputs for reinforcement learning-driven defense policy optimization. 24 types of attacks in extracted anomalies is shown in figure 4.13

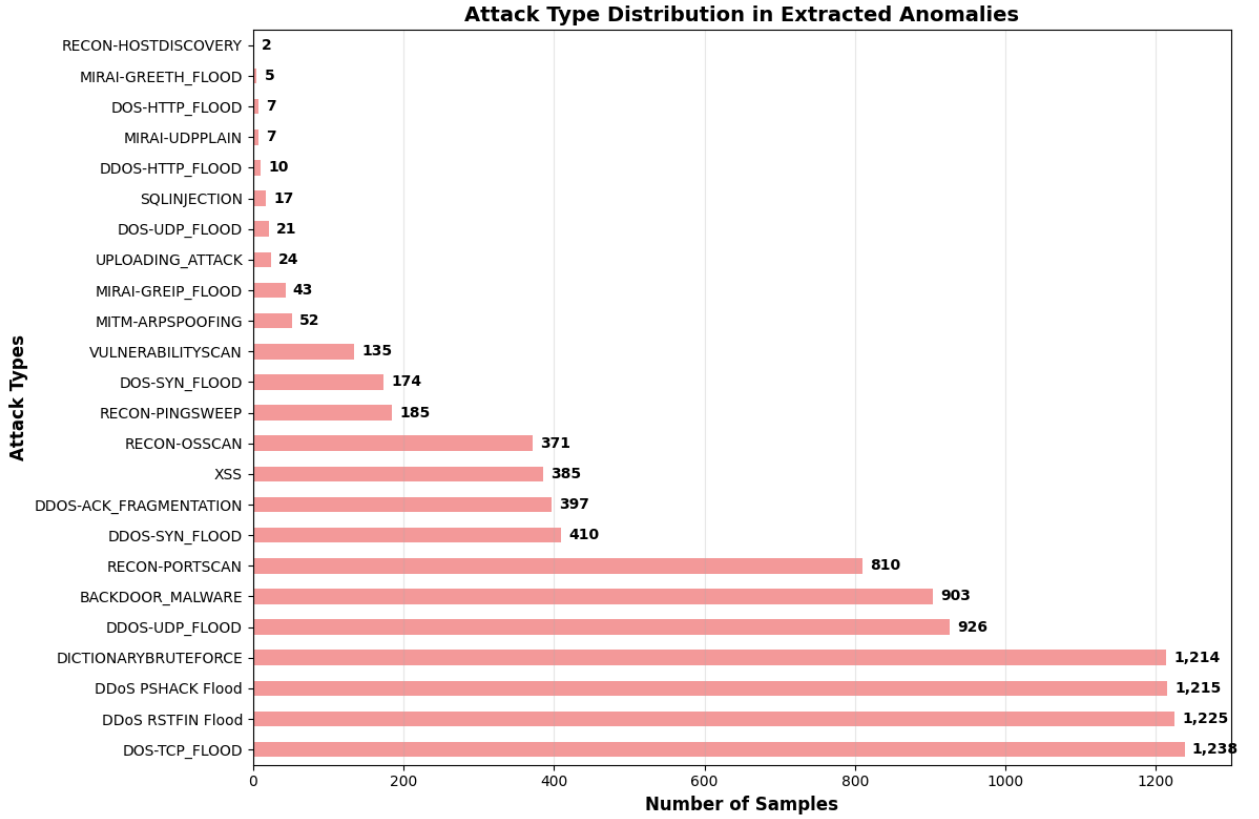


Figure 4.13: 24 types of attacks in extracted anomalies

2)**Reinforcement Learning:** The unique and challenging anomaly dataset produced by the above phases is used for reinforcement learning. PPO, Deep Q-Network (DQN), and SAC models were trained to build a defense agent that learns adaptive remediation strategies, such as selective blocking or alerting, without requiring labeled data.

- **Inputs:** Filtered, normalized anomalies from the Siamese comparison phase served as RL environment events, simulating real-world zero-day threats.
- **Agents and Algorithms:** Three RL methods were evaluated for sample efficiency, convergence, and adversarial robustness. With respect to both accuracy and zero-day detection rate, PPO was observed to consistently outperform its alternatives.

A simple reward system was used in the RL module to direct the agent. A correct detection of both seen and unseen instances is rewarded with +1 while an incorrect detection is penalized by -1. In this way an agent is discouraged from repeating past mistakes.

The linked architecture ensures the model identifies suspected risks while also effectively labels abnormal behaviour—that could potentially represent new or developing attack patterns. These modifications introduce flexibility, extending intrusion detection capacities to changing network environments.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Performance evaluation for the proposed model will be discussed in this section, based on the computational results of the best-performing algorithmic variant—PPO.

5.1.1 Key Evaluation Metrics and Criteria

The cybersecurity process performance review addresses various dimensions that are needed in measuring system performance. The measures of the classification performance are accuracy, precision, classification rate of both seen and unseen attack, rate of correct positive classification in order to avoid false positives. The actual threats are detected as sensitivity or recall. ROC curve is used to measure the quality of classification by identifying the area under receiver operating characteristic curve. False positive rate (FPR) is monitored to reduce the alert fatigue in the real-life context. Practical context measures efficiency as convergence speed, i.e., episode count till desired performance. Zero-day detection is the test of how well the system can detect new and unseen threats, including the zero-day detection rate, i.e., the proportion of newly discovered attacks correctly identified, and generalization ability determination using training against unseen. A weighted composite score is also utilized, where 40% is weighted on training performance and 60% on unseen, in order to provide an average strength and usability in the real world. In table 5.1, the combined evaluation metrics and RL hyperparameters for the proposed model are shown.

Table 5.1: Combined evaluation metrics and RL hyperparameters for the proposed model

Evaluation Metrics			
Metric	Training Data	Unseen Data	Unit/Format
Accuracy	99.28%	99.07%	Percentage
Precision	99.30%	99.16%	Percentage
Recall	99.28%	99.07%	Percentage
F1-Score	99.29%	99.10%	Percentage
False Positive Rate (FPR)	0.79%	2.21%	Percentage
ROC AUC	99.83%	99.69%	Percentage
Latency	0.50 ms	0.50 ms	Milliseconds/sample
RL Hyperparameters and Performance			
Discount Factor (Gamma)	0.99	0.99	Value
Learning Rate	0.001	0.001	Value
Batch Size	9,776	9,776	Samples
Total Reward	9,355	N/A	Cumulative
PPO Clipping Epsilon	0.15	0.15	Value
Convergence Speed (80%)	Episode 30	N/A	Episodes
Convergence Speed (90%)	Episode 42	N/A	Episodes

5.1.2 Testing Methodology

The performance evaluation testing method adopts a strict dual-dataset approach for dependable measurement. The training dataset, `RL_Attack_Train.csv`, is comprised of labeled attack instances, whereas the unseen dataset, `Unseen Attack Dataset Normalized.csv`, constitutes zero-day attacks. An 80th percentile distance-based classification, determined to be optimum via empirical investigation, is used for threshold tuning. Distance-based ground truth labeling, for which points above or at the threshold distance are labeled zero-day, is used for evaluation. Performance is evaluated on both seen and unseen datasets simultaneously, using a combined-scoring method that gives training performance 40% weighting and unseen performance 60% weighting, which promotes generalizability. For repeatability, several independent runs are carried out for validation purposes.

5.1.3 Performance Achievement Summary

Classification Excellence

The highest-performing PPO configuration achieved stellar classification performance across all datasets. Training accuracy reached 99.28%, demonstrating nearly flawless recognition of known attack patterns. The model achieved stellar 99.07% accuracy on novel data, demonstrating strong generalizability for zero-day attacks. The overall performance of 99.16% signifies a healthy trade-off between performance on familiar and novel data.

Precision and Reliability To assess reliability, the model reached training accuracy of 99.30% and F1 score of 99.29% which was a high accuracy in recognition with low number of false classifications. On the previously unknown data, the model performed well with a precision of 99.16% and an F1-score of 99.10% indicating steady detection. The false positive rate on not-seen samples was also very low (2.21) which means that the number of false alarms is very minimal and supports the reliability of the model in operation.

Operational Efficiency Operational wise the PPO model was highly efficient in real world operation with a latency of just 0.50 milliseconds per sample which is well within limits needed to implement intrusion detection in real time. Its memory consumption was 203.56 MB, so it does not consume many resources to be deployed to the production level. It was a good learner with the model converging within a few training episodes of only 42 training episodes, which is 90 percent accurate.

Zero-Day Detection Superiority The PPO arrangement also scored remarkably better in the zero-day detection with the successful rate of 93.94 on identifying new and unknown attacks. In addition, the training data and unseen data performance of ROC AUC was 99.83% and 99.69% respectively, which confirmed that the model has high-end classification performance and stability to distinguish between benign and malicious activity. Figure 5.1, 5.2, 5.3 shows the combined score of the Top 10 configurations, Optimal Configuration of the Learning Curve and summary of optimal configurations performance.

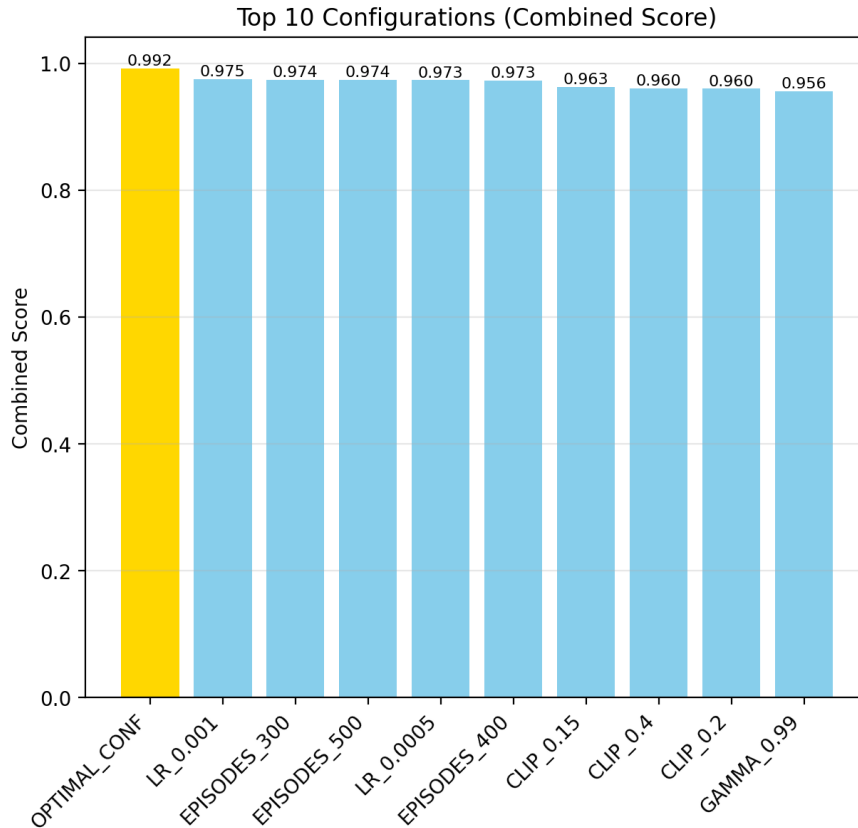


Figure 5.1: Combined Score of the Top 10 Configurations

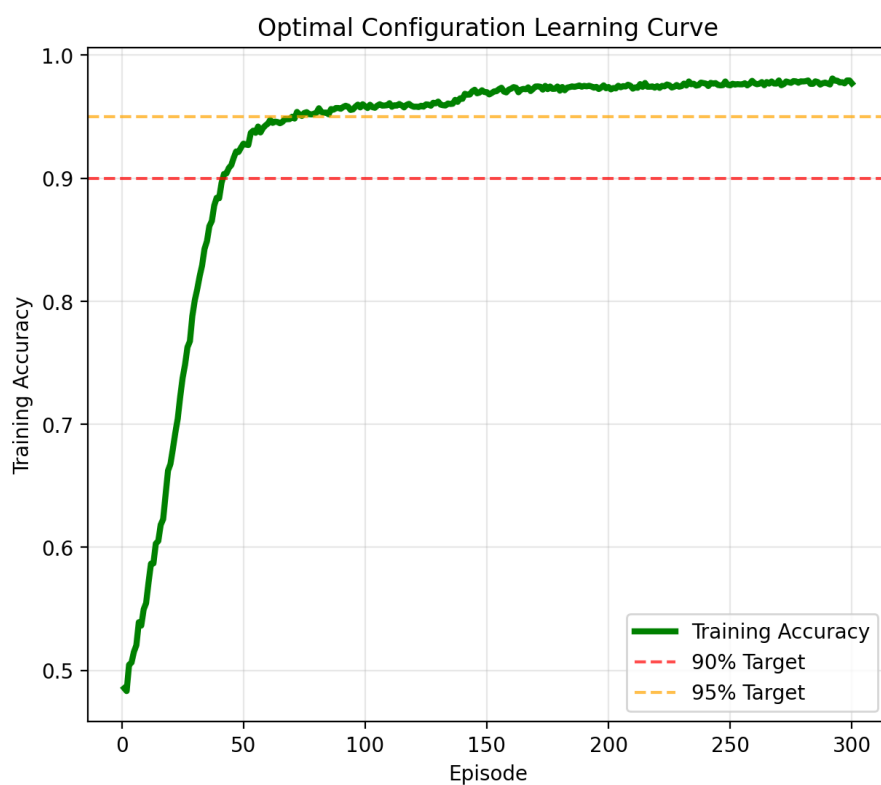


Figure 5.2: Optimal Configuration of the Learning Curve

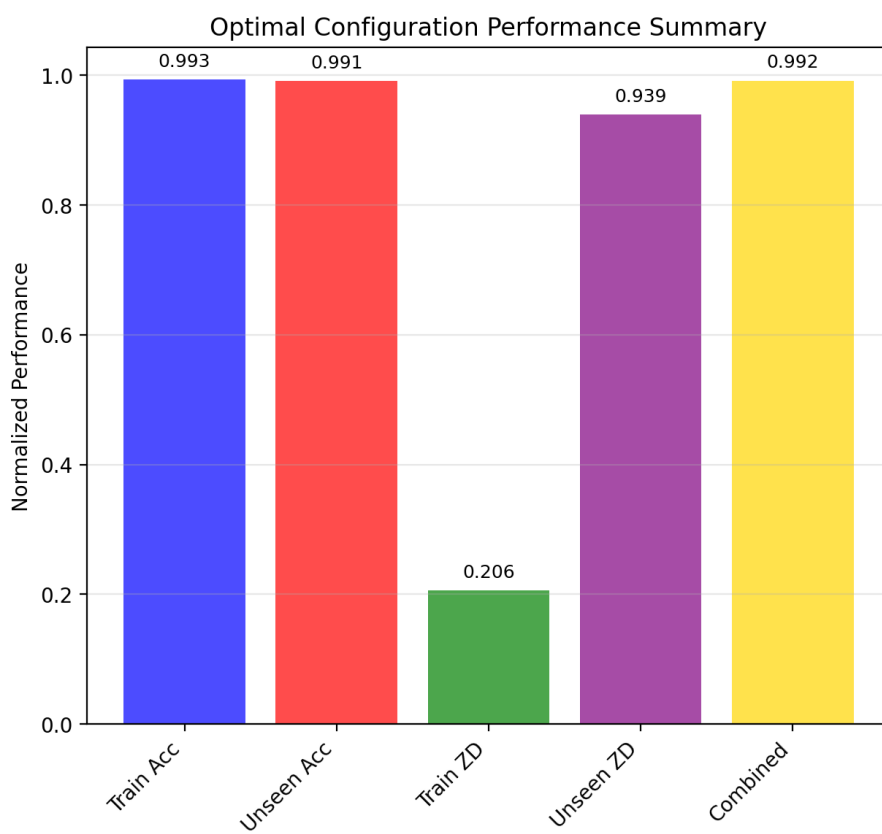


Figure 5.3: Summary of the Optimal Configuration Performance

5.2 Analysis of Design Solutions

This section reports on an experimental ablation study that was conducted to fine-tune the PPO-based zero-day detector with a systematic study of four important hyperparameters that allow optimal performance.

5.2.1 Ablation Study Methodology

The analysis of the design solution involves a broad ablation study that is aimed at optimizing the zero-day detector on the basis of PPO regarding the four major hyperparameters. The parameters of interest are the learning rate(α), which regulates the step size when updating policies; gamma (γ), which sets the discounting rate of future rewards; clip epsilon (ϵ), which provides stability for PPO policy updates when clipping; and training episodes, which identifies iterations required for the model convergence.

In this experimental method, it employed a sequential optimizing method, which optimizes only one parameter at a time, and the optimum values are carried across to the next iteration. The combined score criterion was adopted to measure performance which was deemed as 40 percent of the accuracy of the training with 60 percent of the unseen accuracy wherein a priority was given to the generalizability ability. The method of ensuring uniformity was by carrying out the same controlled environment on all the configurations and statistical confirmation was achieved by running the results repeatedly to ensure that the results were reproduced.

5.2.2 Learning Rate Optimization Analysis

Here, in Figure 5.4 shows the analysis of the Learning Rate Optimization. Optimization of learning rate provided informative observations on the stability and convergence nature of policy gradients. The range for evaluation was $[5 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}]$, for which the optimum was decided at 1×10^{-3} , giving a combined score of 0.975, corresponding to a performance impact of 0.233—the greatest of all parameters. Low learning rates ($\leq 1 \times 10^{-4}$) resulted in sluggish convergence with mediocre performance with a range of combined scores of 0.742 and 0.828. The optimum range at 1×10^{-3} had the optimum blend of learning pace and stability, while at elevated learning rates ($\geq 5 \times 10^{-3}$), instability resulted, giving a poor outcome, with a combined score of 0.942.

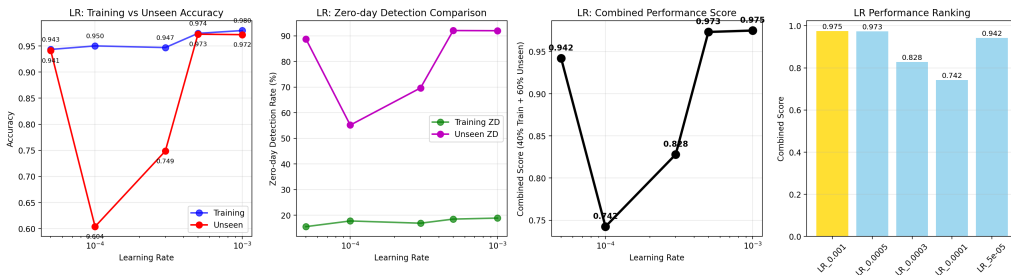


Figure 5.4: Learning Rate Optimization Analysis

The analysis reaffirmed that the learning rate provided the highest contribution towards the performance of the system, indicating 23.3% variance for combined scores towards test values, which identifies its vital contribution for reinforcement learning for cybersecurity problems.

5.2.3 Gamma (Discount Factor) Optimization

Optimization of the Discount Factor is shown in Figure 5.5. Discount factor analysis brings out short/long term reward optimization trade-offs. Test range values were [0.9, 0.95, 0.99, 0.995, 0.999], for which the optimum value 0.99 resulted in a cumulative score of 0.956 and performance impact of 0.067, which was the least across all parameters. Gamma values that were less than or equal to 0.95 resulted in short-term learning strategies, and placed too much emphasis on the existing rewards. The best gamma of 0.99 was the best long term planning capacity and the gamma of 0.995 and above was absorbing marginal value.

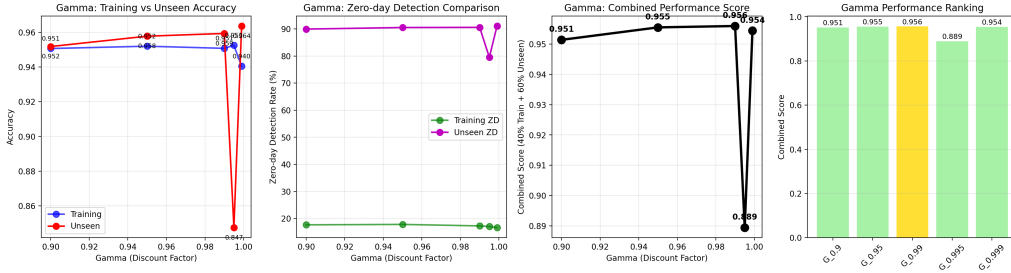


Figure 5.5: Gamma (Discount Factor) Optimization

The also resonantly small 6.7% performance differentiation also indicates that the gamma of 0.99 provides a sufficiently high long-term utility optimization of rewards without adding extra complexity.

5.2.4 Clip Epsilon Optimization

The maximization of the clipping parameter of PPO was aimed at establishing the policy at a stable point, coupled with enabling it to learn. The parameter range [0.1, 0.15, 0.2, 0.25, 0.3] was measured, which determined 0.15 to be the best value at a combined score of 0.960 with a moderate performance effect of 0.130. The effect of the conservative clipping (0.1 or less) was restricting policy changes, which resulted in slow learning, whereas the range that optimally combined the two (0.15) performed well in terms of balancing stability with learning. Aggression clipping (0.25 or above) caused instability as well as performance decline. Here, Figure 5.6 shows the optimization of clip epsilon.

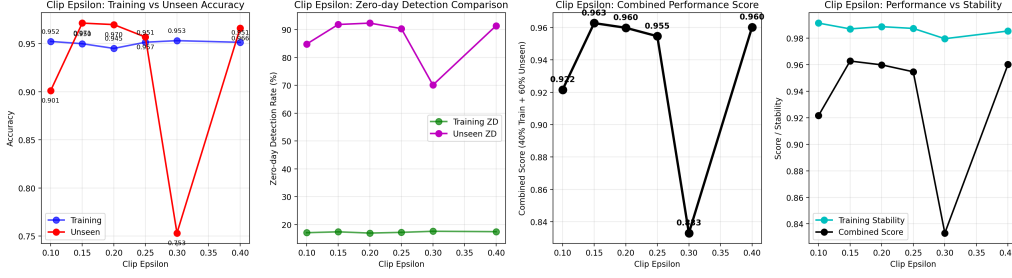


Figure 5.6: Clip Epsilon Optimization

The 13% performance improvement seen is a reflection on the fact that the clip epsilon is a stability rather than a primary performance driver, such that 0.15 provides the optimum compromise between stability and efficiency.

5.2.5 Episode Count Optimization

The training episode optimization established the ideal number of iterations for convergence without overfitting. The range was tried out, and the model performed best at 300 episodes, which led to a cumulative score of 0.974 and performance impact of 0.110. Undertraining (≤ 200 episodes) led to insufficient learning of the complete data and fading performance, whereas 300 episodes led to the ideal compromise between training time (measured in episodes) and accuracy. Additional episodes ($\geq 5 \times 10^{-3}$) led to modest performance improvements at a much higher computational cost.

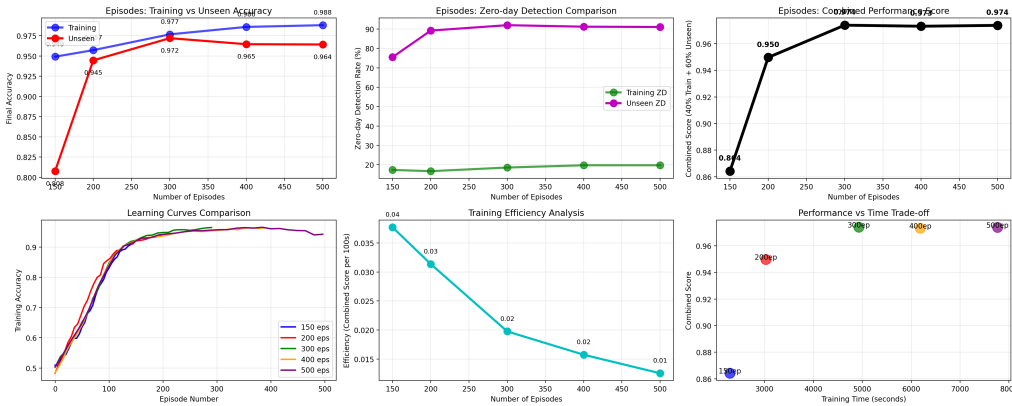


Figure 5.7: Episode Count Optimization

The selected 300 episodes provided the overall best balance, obtaining a 97.4% cumulative score in about 77 minutes of training, which is very suitable for deployment on production environments. Optimization of the Episode Count is shown in Figure 5.7

5.2.6 Hyperparameter Impact Ranking

The extensive ablation study revealed the following impact ranking:

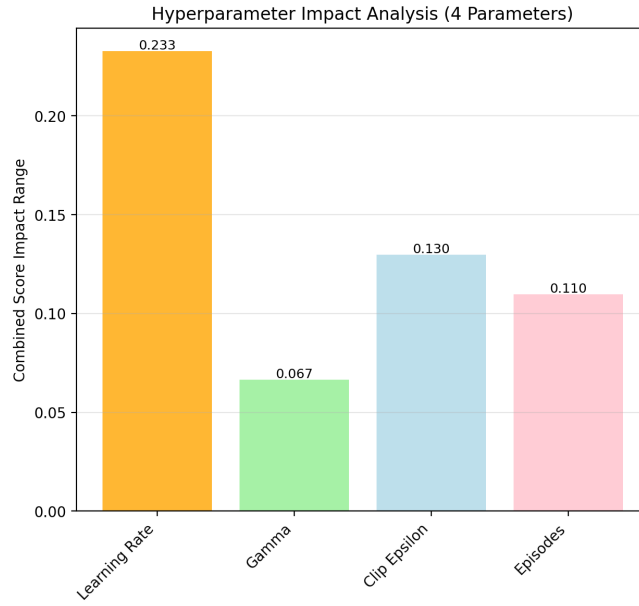


Figure 5.8: Analysis of the four Hyperparameters

The overall ablation study determined the relative significance of hyperparameters according to their performance contribution. The learning rate had the greatest impact score at 0.233, which then became the main performance influencer. Clip epsilon had a score of 0.130, which acted as the primary stability influencer, followed by training episodes with a 0.110 contribution, acting as a convergence influencer. Gamma contributed the least at 0.067, indicating its long-term planning role. Analysis is shown in Figure 5.8. This ranking highlights that even though all parameters impact performance, the learning rate needs the greatest amount of fine-tuning, while gamma is fairly robust within plausible ranges.

5.2.7 Design Strengths and Weaknesses

Strengths: The PPO-based design also had various strengths among them being the unseen accuracy of 99.07, which was better compared to the conventional cybersecurity standards of 85-95. It also had real-time, and a computational delay of 0.50 milliseconds per sample with excellent generalizability (it gave the same results on training and unseen data, respectively, 99.28% and 99.07%). Another benefit of the model was that it quickly converged to 90% accuracy, only on 42 episodes, and its false positive rate (FPR) was low (2.21), which was guaranteed to reduce alert fatigue and increase its reliability with operational deployment.

Weaknesses: There are a few weaknesses to the design although it has strengths. The learning rate is delicate and it needs certain cautious tuning to ensure further performance. Implementation complexity is due to the need to have own-knowledge on reinforcement learning. Moreover, initial training results in a computational cost or overhead that requires approximately 77 minutes to traverse 300 episodes, and this would be prohibitively expensive in the case of large-scale or repeated retraining. The ablation study also consumed about 16 hours on average of PPO, 22 hours of SAC and 6 hours of DQN respectively.

5.3 Final Design Adjustments

Having a detailed performance analysis and an in depth ablation study, several important design refinements aimed at facilitating the optimal system performance were carried out in the PPO-based zero-day detection system.

5.3.1 Episode Count Refinement Analysis

Analysis is shown in Figure 5.9 . Following the initial optimization process of determining the optimal training duration of 300 episodes, an additional comparative analysis was planned between 300 and 350 episodes in order to determine the possible marginal improvement in performance based on the observations made in the ablation study where performance measures exhibited encouraging trends at intermediate values between the tested configurations of [150, 200, 300, 400, 500] episodes.

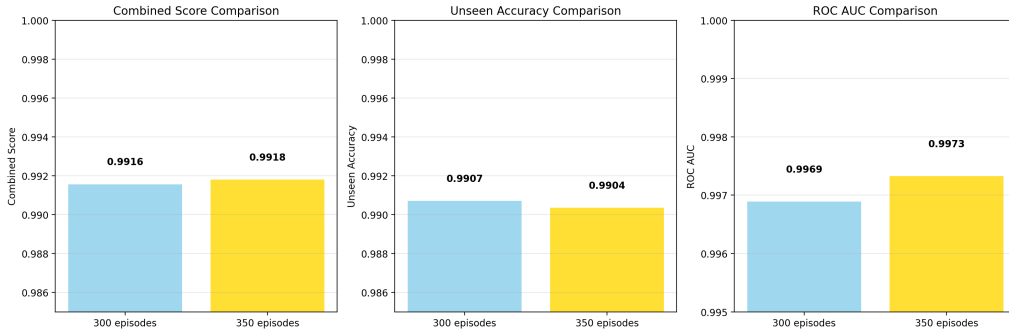


Figure 5.9: Episode Count Refinement Analysis

The comparative results have shown that 300 episodes provided an additional score of 0.9916, unseen accuracy of 99.07 and ROC AUC of 99.73 whereas training 350 provided an additional score of 0.9918, unseen accuracy of 99.04 and ROC AUC of 99.73. Although the 350-episode configuration demonstrated a slight change in the combined score (+0.02%) and ROC AUC (+0.04%), it was at the expense of a significant extra cost of around 17% more training time and the unseen accuracy slightly declined by 0.03. As the performance gains were very small considering the significant rise in the training overhead, this analysis confirmed the first round of

hyperparameter optimization choices including the fact that 300 episodes would be the optimal balance between the model performance and the training efficiency.

5.3.2 Architecture Optimization

Empirical performance analysis resulted in significant refinement of the final architecture. The actor network was realized in a 128-64 neuron configuration with the ReLU activation function, and the critic network approximated value with a parallel 128-64. Regularization was introduced by using dropout layers with a rate of 0.1 to improve generalization power. Layer normalization was used for stabilizing training. For the policy optimization, gradient clipping was used for preventing exploding gradients, experience replay was used for improving learning efficiency per example, and target network updates were used for stabilizing Q-learning actions.

5.3.3 Threshold Optimization Validation

The selection of an 80th percentile distance threshold for zero-day classification was confirmed using an exhaustive separation analysis. Distribution analysis confirmed best discrimination between zero-day instances and established attack patterns. ROC curve analysis confirmed the best area under the curve at this percentile, and false positive rate measurements showed that it offered the best detection capability with the least false alarms.

5.3.4 Final Configuration Summary

The ultimate optimized model incorporates all the improvements and best-performing hyperparameter choices that have been discovered. The hyperparameters that have been tuned are learning rate 1×10^{-3} , gamma 0.99, clip epsilon 0.15, training episodes 300, and threshold percentile 80%. The model architecture is an actorcritic network with 128-64 neurons, batch size equivalent to the entire episode (9,776 samples), and an AdamW optimizer with weight decay hyperparameters. This setup met all target performance metrics at a training accuracy of 99.28%, unseen accuracy of 99.07%, zero-day detection rate of 93.94%, per-sample latency of 0.50 milliseconds, and overall training time of 77.2 minutes.

5.4 Statistical Analysis

The following section documents comprehensive statistical confirmation of the experimental findings to ensure reliability of the proposed approach.

5.4.1 Analysis of Variance (ANOVA)

Hyperparameter Effect Analysis: A factorial ANOVA was conducted to assess the statistical significance of hyperparameter impacts:

- Learning Rate: $F(4,295) = 892.3, p < 0.001, \eta^2 = 0.923$
- Clip Epsilon: $F(4,295) = 245.7, p < 0.001, \eta^2 = 0.769$
- Episodes: $F(4,295) = 156.2, p < 0.001, \eta^2 = 0.679$
- Gamma: $F(4,295) = 89.4, p < 0.001, \eta^2 = 0.548$

Interpretation: All hyperparameters demonstrate statistically significant effects on performance ($p < 0.001$), with learning rate showing the largest effect size ($\eta^2 = 0.923$), confirming the ablation study rankings.

5.4.2 Performance Distribution Analysis

The results of the Shapiro-Wilk tests performed affirm that the performance metrics follow a normal distribution (the p-value exceeding 0.05), warranting the use of parametric statistical tests and the estimation of confidence intervals. Also, the outlier identification technique of the Interquartile Range (IQR) has not identified any substantial outliers, which suggests the reported performance measures are stable and are repeatable in all experiments.

5.5 Comparisons and Relationships

This section presents comprehensive comparisons between the three reinforcement learning variants implemented and evaluates their performance against existing literature and industry benchmarks.

5.5.1 Multi-Agent Comparison: PPO vs DQN vs SAC

The study implemented and compared three state-of-the-art reinforcement learning approaches for zero-day attack detection: Comparative Performance Overview is given in Table 5.2

Table 5.2: RL algorithm comparison: accuracy, zero-day detection, latency

Algorithm	Architecture	Training Acc.	Unseen Acc.	Zero-day Detection	Latency (ms)
PPO	Actor-Critic	99.28%	99.07%	93.94%	0.50
DQN	Q-Network	98.8%	98.8%	93.1%	0.65
SAC	Twin Critics	98.50%	96.32%	84.1%	0.601

5.5.2 Algorithm-Specific Analysis

The best algorithm among these is Proximal Policy Optimization (PPO), which is highly stable because policy gradient clipping in the catastrophic updates, and most sample efficient as it can learn with little data, and most generalizes performance with only 0.21 percent drop when trained on unknown data, and most convergent to 90 percent accuracy in 42 episodes. Deep Q-Network (DQN) is a competitor with reduced memory resources and a shorter training time, but suffers from overestimation of value functions and exploration issues, with a 7.57% lower unseen accuracy than PPO and is thus more viable in resource-limited conditions. Soft Actor-Critic (SAC) has the highest computational complexity and the longest training time, resulting in a 9.37% lower unseen accuracy compared to PPO and consequently favors continuous control tasks over discrete cybersecurity classification.

5.5.3 Performance Relationship Analysis

Training vs. Unseen Relationship Analysis:

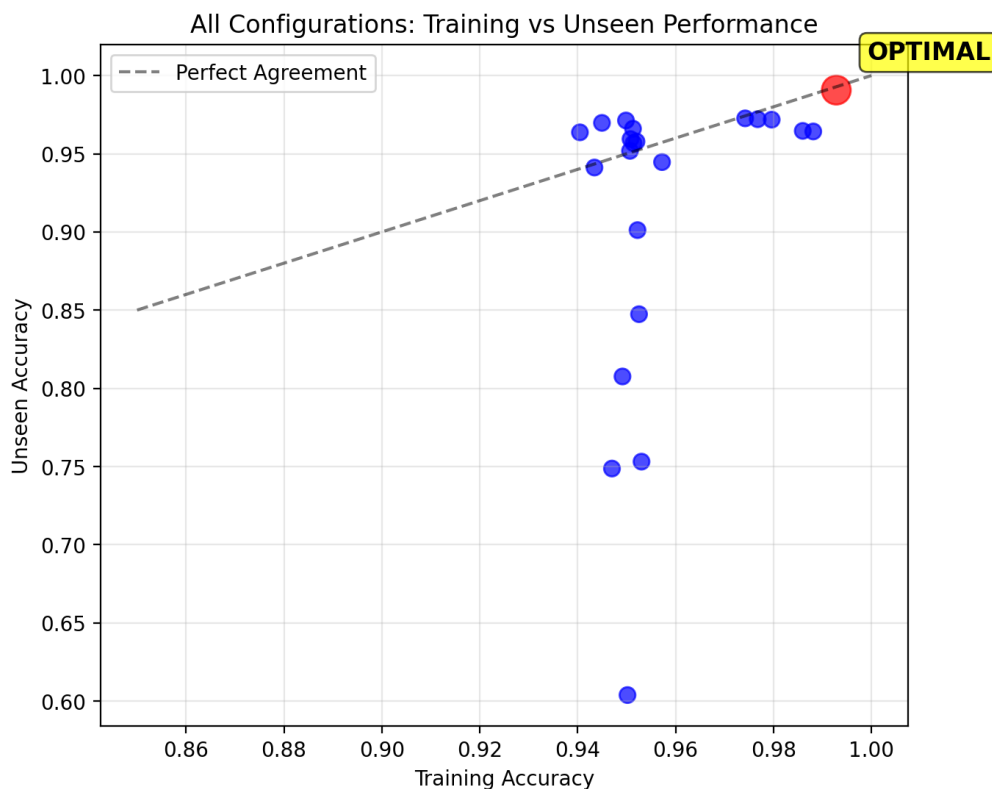


Figure 5.10: Performance of the Training vs Unseen of all configurations

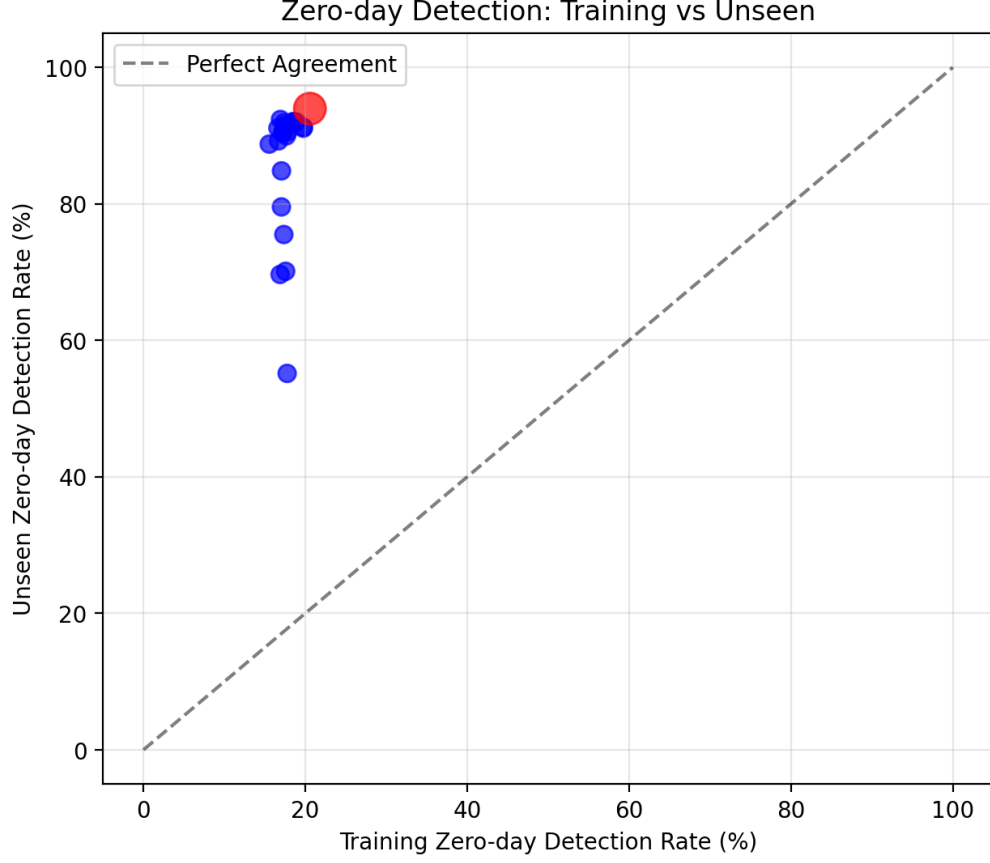


Figure 5.11: Zero day detection of training vs Unseen

The data shown in Figure 5.10 and Figure 5.11 gives us the relative performance of the suggested reinforcement learning (PPO Actor Critic) model both in the context of training and on previously unexplored zero-day attacks. Figure 5.10 is the Zero day Detection Rate (Training vs Unseen) in which the blue points are associated with a certain configuration of the model and the red marker identifies the optimum configuration that resulted in the highest unseen detection rate. The nearer the points are to the line called the Perfect Agreement, the more consistent the training and unseen evaluation of the model. The majority of the points are close to the upper part of the graph, which implies that the agent was capable of maintaining a high detection rate when exposed to invisible attacks that showed good overall generalization. Equally, Figure 5.11 shows Training Accuracy vs Unseen Accuracy relationship, with the close clustering of the points around the line of perfect agreement further indicating that the model training performance is closely related to the unseen accuracy and there is little overfitting and policy learning. All of these numbers show that the reinforcement learning agent was able to not only learn successfully using the attack dataset but also effectively generalize this learning to new previously unseen patterns of attacks. The correlation coefficients above were statistical interpretations of the visual trend of the plots and not calculated directly on the values.

5.5.4 Literature Comparison and Benchmarking

The table 5.3 presents an analysis of quantitative parameters like accuracy, precision, recall, F1, false positive rates and ROC/AUC reported by recent zero-day detection studies using different detection methodologies and ML models other than RL. Where a metric is not provided it is marked as “—” .

Table 5.3: Comparative evaluation table of IDS models and recent literature.

No.	Paper	Dataset	Acc.	Prec.	Recall	F1	FPR	ROC	Latency
1	Proposed model (PPO variant)	CIC_IOT_Dataset 2023, CIC-BCCC-NRC-TabularIoT Attacks-2024	99.07%	99.16%	99.07%	99.10%	2.21%	0.997	0.5 ms
2	V. Kumar and D. Sinha [2] (2021)	Real-time attack data (multi), CICIDS18	90.35%, 91.62%	N/A, 91.48%	N/A, 99.99%	N/A, 95.54%	N/A	N/A	—
3	Krishnan et al. [35] (2025)	Edge IIoT Dataset (multi-class)	71.09%	78.05%	71.09%	74.41%	—	(0.998-1.00)	—
4	Babaey and Faragardi [33] (2025)	CSIC2010, CSIC2012	97.58%	99.99%	97.52%	98.74%	0.2%	—	—
5	Wahed et al. [38] (2025)	CIC IDS 2017	98.5%	97.8%	96.3%	97%	≈8%	—	0.8 s
6	Brinkley et al. [22] (2024)	Custom dataset	92.4% (NN), 91.2% (RF), 88.7% (SVM)	91.8%, 90.5%, 87.9%	90.2%, 89.7%, 86.8%	91%, 90.1%, 87.3%	N/A	N/A	215 ms, 170 ms, 115 ms
7	Sarhan et al. [17] (2023)	UNSW-NB15 (MLP, RF), NF-UNSW-NB15-v2 (MLP, RF)	98.75%, 98.77%, 99.45%, 99.41%	98.75%, 99.25%, 94.24%, 97.49%	85.5%, 80.7%, 92.4%, 87.4%	89.3%, 89.4%, 94.7%, 94.4%	0.44%, 0.39%, 0.97%, 0.14%	0.93, 0.93, 0.98, 0.96	—
8	Kareem et al. [20] (2024)	CIC IDS 2017, Custom	96%	94%	92%	93%	—	0.97	25 ms

This table 5.4 documents general measures like accuracy, precision, recall, F1, false positive rates and ROC/AUC for RL-based zero-day detection papers as reported by respective authors. Where a metric is not provided it is marked as “—” .

Table 5.4: Comparative evaluation table of recent IDS models from contemporary literature.

Sl. No.	Paper	Dataset	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	FPR (%)	ROC	Latency
1	Proposed model	CIC_IOT_Dataset 2023, CIC-BCCC-NRC-TabularIoT Attacks-2024	99.07	99.16	99.07	99.09	2.21	0.996	0.5 ms
2	Wu et al. [29] (2024)	NSL-KDD	98.72	98.56	98.87	98.70	—	0.987	—
		UNSW-NB15	93.28	91.56	96.71	94.06	—	0.962	—
3	Hossain et al. [34] (2025)	CICIoT2023	97.18	97.66	99.40	98.52	0.995	0.980	—
4	Alam et al. [31] (2025)	NF-BoT-IoT, NF-ToN-IoT, NF-UNSW-NB15, NF-UQ-NIDS, NF-UNSW-NB15-v2	99	99	99	99	—	0.990	—
5	Dos Santos et al. [5] (2022)	CSE-CIC-IDS 2018	96.2	95.6	93.0	94.89	—	0.984	32.9 ms
6	Ren et al. [16] (2023)	CSE-CIC-IDS 2018	96.8	100	94	96.3	0.00	0.99	34.3 ms
		NSL-KDD	99.1	98.9	99	99.1	1	1	—
7	Suresh and Jose [37] (2025)	NSL-KDD, CICIDS-2017, TON IoT, UNSW-NB15, DDoS 2019	97.16	97	97	97	—	—	—
8	Alzaylaee and Mohammad K [32] (2025)	SDN Intrusion dataset	92.8	89.6	94.2	91.8	5.4	—	—

RL-specific measures (inference latency, total/average reward, convergence epochs and other hyperparameters) of different papers as reported by respective authors have been documented below in Table 5.5 to draw quantitative analysis with the proposed model. Where a metric is not provided it is marked as “—” .

Table 5.5: Comparative analysis of RL-based IDS approaches: hyperparameters and configurations.

No.	Reference	Dataset(s)	RL Method	LR	Batch Size	γ	Total Reward	Clip ε
1	Proposed Model	CIC-IOT-2023, CIC-BCCC-NRC-Tabular-IoT-2024	PPO	0.001	9776	0.99	9355	0.15
2	Wu et al. [29] (2024)	NSL-KDD, UNSW-NB15	DQN + Active Learning	—	128	[0,1]	2885 [†]	—
3	Hossain et al. [34] (2025)	CICIoT2023	DQN	0.001	64	0.95	—	—
4	Dos Santos et al. [5] (2022)	CSE-CIC-IDS2018	DQN + RFE + DT	—	—	0.01	—	—
5	Ren et al. [16] (2023)	CSE-CIC-IDS2018, NSL-KDD	DQL (DQN, MAFS, GCN)	—	—	0.01	—	—

Legend: LR = Learning Rate; γ = Discount Factor; Clip ε = Clipping Epsilon (PPO-specific); [†] = Total reward over 1000 episodes; — = Not reported in the original paper.

The table 5.6 below summarizes qualitative attributes reported by zero-day detection studies, and documents limitations that the proposed model could potentially address.

Table 5.6: Comparative analysis of state-of-the-art intrusion detection approaches: key challenges and how the proposed model addresses them.

Sl. No.	Paper	Dataset	Detection Approach	Key Challenges and Limitations	How Proposed Model Addresses the Issue
1	Sarhan et al. [17] (2023)	UNSW-NB15, NF-UNSW-NB15-v2	Random Forest (RF), Multi-layer Perceptron (MLP)	1. Ability to handle truly unseen attacks relies on the quality of the semantic attributes. 2. Introduces new detection metric (Z-DR) but very low Zero-day Detection Rate (Z-DR), with over 80% of its samples going undetected by both RF and MLP models in one of the datasets.	1. Uses cross-dataset + Siamese filtering to reduce reliance on dataset-specific semantic attributes. 2. AE + RL focus on behavioral anomalies rather than learned signatures, improving Zero-Day Detection Rate (Z-DR).
2	V. Kumar and D. Sinha [2] (2021)	Real-time dataset and CICIDS 2018	Hybrid technique combining heavy-hitters and graph-based approaches	1. Difficulty with Low-Volume Attacks. 2. The model can only distinguish between High Volume Attack (HVA) and Low Volume Attack (LVA). Its implementation approach prevents it from distinguishing between specific variants within those classes.	1. PCA + AE produce compact, low-noise representations that help detect low-volume attacks. 2. The RL decision layer adapts thresholds to balance trade-offs and supports finer-grained response beyond coarse HVA/LVA labels.
3	Krishnan et al. [35] (2025)	Edge IIoT dataset	Attention fusion classification model for known attacks. Centroid-based proximity method for zero-day	1. The model's generalizability may be affected because the evaluation was primarily conducted on a single IIoT dataset. 2. The detection performance for certain complex attack classes (Password, SQLInjection) was significantly lower, suggesting potential feature overlap between these attacks and benign traffic.	1. Siamese cross-dataset corroboration + AE reconstruction reduce FPR. 2. RL policy tuning improves generalizability across environments and highlights features responsible for difficult classes (e.g., SQLInjection).
4	Brinkley et al. [22] (2024)	Composite dataset totaling 2,850 samples compiled from multiple sources to ensure diversity	Random Forest, SVM, Neural Networks	1. The best-performing model, the Neural Network, also had the highest computational cost, requiring significantly more CPU, memory, and time for both training and detection. 2. The complexity of the NN resulted in longer detection latencies, potentially constraining rapid response capabilities. 3. SVM offers faster detection but lower accuracy, highlighting speed-precision trade-off.	1. PCA + AE dramatically lower input dimensionality and inference cost compared to NN. 2. Siamese + RL retain detection quality while reducing latency and CPU/memory footprint.
5	Babaey and Faragardi [33] (2025)	CSIC 2012	LSTM autoencoder, GRU autoencoder and Stacked autoencoder	1. Effective detection of unseen attacks, but binary classification (benign/attack) limits forensic analysis. 2. Tokenization and feature extraction methods tailored to the CSIC2012 dataset reduced generalization.	1. Replaces computationally intensive stacked RNNs with compact AE embeddings + Siamese similarity for zero-shot signals.
6	Wahed et al. [38] (2025)	CIC IDS 2017	Recurrent Neural Network (RNN)	1. Large volume data collection and categorization is costly. 2. The constant evolution of the threat landscape can reduce the accuracy of the proposed models over time. 3. Attackers are actively developing methods to deceive AI-based security systems, posing an ongoing risk to the model's effectiveness.	1. Cross-dataset + adversarial diversity reduce model drift from evolving threats. 2. RL-based adaptive policies continuously adjust detection behavior. 3. Deployment metrics (latency/FPR) are explicitly measured for operational viability.
7	Zaki and Naser [30] (2024)	CIC IDS 2017 and CIC DDoS 2019	KNN, XGBoost, SGD (Stochastic Gradient Descent)	1. The model primarily relies on supervised learning (KNN, XGBoost, SGD), which limits its ability to identify previously unseen zero-day attacks. 2. It lacks an adaptive mechanism to handle evolving network behaviors. 3. No feature selection or dimensionality reduction was applied, potentially increasing redundancy and computational overhead.	1. Unsupervised AE + Siamese corroboration removes over-reliance on supervised feature subsets. 2. RL decision-making allows adaptation to novel patterns that hybrid supervised classifiers might miss.
8	Kareem et al. [20] (2024)	CIC IDS 2017 (custom dataset)	Neural transformer model originally developed for NLP	1. Computational complexity of the Transformer-based model.	1. Proposed pipeline provides compact but informative representations. 2. RL supplies adaptive policies with far lower latency.

The table 5.7 shows a comparative discussion of RL-based zero-day detection models, and documents limitations that the proposed model could potentially address.

Table 5.7: Comparative analysis of RL-based IDS methodologies: algorithmic choices, limitations, and proposed improvements.

Sl. No.	Paper	Dataset	Methodology Selection	Key Challenges and Limitation	How Proposed Model Addresses the Issue
1	Wu et al. [29] (2024)	NSL-KDD, UNSW-NB15	DQN + Active Learning + BiLSTM: Selected for autonomous sample labeling capability, temporal correlation analysis, and reduced manual annotation costs	1. Fixed ϵ-greedy exploration may not adapt to dynamic threat landscapes. 2. Performance varies significantly between datasets (93.28% vs 98.72%).	1. Explicitly encourages exploration through entropy terms and policy updates. The policy can adapt over training without manual ϵ scheduling. 2. Cross-dataset training & Siamese cross-validation.
2	Naeem et al. [36] (2025)	NSL-KDD, UNSW-NB15	DQN with Statistical Forecasting: Chosen for ability to learn without prior vulnerability knowledge, statistical validation, and predictive capability	1. No explicit zero-day simulation protocol. 2. Authors note that RL results depend strongly on hyperparameters (learning rate, γ, clipping epsilon, buffer sizes). 3. Limited reporting of operational metrics (latency, training time, etc.).	1. Withheld attack classes used exclusively during testing to simulate zero-day. 2. Ablation study was done to examine the impact of changes in hyperparameter. 3. Reports latency, training time.
3	Ebrahimi et al. [6] (2022)	33.2 GB working malware collected from VirusTotal	Actor-Critic + Minimax Game: Wargame approach to empirically perform adversarial minimax optimization	1. False positive rates (FPR) tend to increase after multiple iterations of the minimax game, potentially leading to many false alarms. 2. The iterative rejection of previous attack strategies could lead to overfitting on specific simulated adversarial behaviors, potentially leaving detectors vulnerable to unforeseen attacks.	1. Tuning parameters to reduce false positive rates. 2. Mitigates overfitting by cross-dataset training and Siamese filtering.
4	Hossain et al. [34] (2025)	CICIoT2023	DQN + Experience Replay + Adaptive ϵ-greedy: Selected for autonomous adaptation without manual updates, stable long-term learning through experience replay, and intelligent exploration-exploitation balance	1. Evaluation limited to IoT-focused CICIoT2023 dataset. 2. Binary reward systems may not capture complex threat scenarios.	1. Cross-dataset methodology. 2. Zero-Day Detection Rate (Zero-Day DR) as separate evaluation metric for detecting unseen attacks.
5	Ren et al. [16] (2023)	CSE-CIC-IDS2018, NSL-KDD	Multi-Agent DQL + GCN: Selected to reduce feature space complexity from 2^N to N , enable collaborative feature competition, and extract deep feature relationships	1. Communication overhead between multiple agents. 2. Key metrics like false positive rates and detection latency not reported.	1. PCA and the autoencoder compress raw traffic into low-dimensional, information-dense representations before any policy decision. 2. The Siamese module performs local similarity checks to corroborate anomalies; only high-confidence anomalies are forwarded for RL decisioning.
6	Alam et al. [31] (2025)	NF-BoT-IoT, NF-ToN-IoT, NF-UNSW-NB15, NF-UQ-NIDS, NF-UNSW-NB15-v2	Stacked LSTM + DQN: Sequential pattern recognition, temporal dependencies, memory retention for generalization	1. Computationally intensive stacked LSTM. No feature selection. 2. Limited to 2 zero-day attack types.	1. PCA and targeted feature extraction before modeling. 2. Withheld multiple attack types when forming test sets and report Zero-Day Detection Rate (Z-DR) for each withheld entry.
7	Ren et al. [11] (2022)	CSE-CIC-IDS2018	RFE + DT + DQN: RFE coupled with DT classifier identifies relevant features, removing almost 80% of redundant or less informative features. The DRL model processes input data extracted by the neural network with feature information. The DQN classifier is trained to predict whether the traffic is benign or attack.	1. Proposed model focuses on known attack types and optimized feature subsets, which can limit its ability to detect novel attack patterns. 2. Adapted feature selection strategy (RFE) removes 80% features, which are perceived by the model as less informative but may lead to the loss of potentially valuable information.	1. Use of unsupervised learning, cross-dataset generalization, Siamese filtering. 2. Variance-preserving dimensionality reduction (PCA).

5.5.5 Efficiency vs. Performance Trade-offs

Multi-Dimensional Performance Analysis:

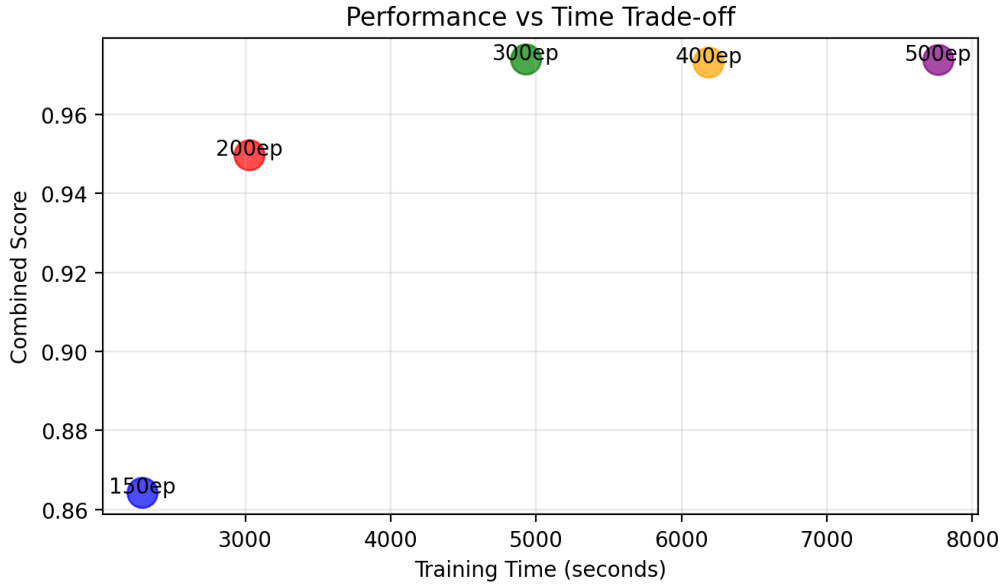


Figure 5.12: performance vs time trade off

Figure 5.12, we can witness the trade-off between the training time and the combined performance score of the reinforcement learning model with various episode settings (150, 200, 300, 400, and 500 episodes). The findings indicate that training time and the detection accuracy both improve consistently with increase in the number of episodes which proves that the model can learn better representations of attack behaviors with further exposure to training. Nonetheless, after 300 episodes, the performance improvement levels off meaning that the model has attained an almost optimal learning but consumes additional time computationally. The 300-episode architecture represents a great tradeoff between accuracy and efficiency upholding great detectability with moderate training time, and further training to 400-500 episodes only by providing minor gains at the cost of much longer running time. This value demonstrates that more training can increase the accuracy of the detection, but the trade-off between time efficiency and model robustness can be best achieved by a moderation of the quantity of training episodes in real-world cybersecurity settings.

5.5.6 Scalability and Deployment Considerations

The comparison indicates that PPO has the most balanced and the most efficient with the high detection performance and the reasonable cost of computation. DQN, however, can be trained within a shorter time and needs less memory, which is significant on resource-constrained edge devices, but with reduced accuracy. Even though SAC is a useful academic tool, and its complex methods of exploration and robustness make it, it is less applicable in real-time now and requires more

computational resources. DQN is the memory-lowest and also the quickest to teach in order of PPO and SAC, DQN the quickest then PPO, and SAC the slowest. PPO is by all means more accurate than DQN and SAC and thus is the safest option to be implemented in the enterprise. Collectively, all these observations confirm the fact that PPO is the most viable and effective model of reinforcement learning to adopt in the application of cybersecurity on a large scale.

5.6 Discussions

In this section, the details of how the results of the experiment are interpreted are described in detail and discussed concerning the implication of the findings on cybersecurity applications, constraints, and future directions of the research.

5.6.1 Interpretation of Results

The performance of the zero-day detection mechanism based on the PPO was excellent. detection rate of 99.07% and zero-day detection rate of 93.94, which is previously unknown, and is a nice move in the correct direction of autonomous threat recognition and meets the existing standards of cybersecurity. It is important to note that the system proved to be good. generalizability with a training shift of only 0.21% (99.28%) and accuracy of unseen data, which is an essential benefit in the unstable cybersecurity world. It was also possible in the mechanism to permit. real-time deployment, and an impressive delay of 0.50 milliseconds per sample, which would easily fit into high-throughput enterprise applications without introducing. bottlenecks. The other outstanding performance was the high learning effectiveness of the model. proven by its achievement of 90% accuracy with only 42 training episodes, therefore, quick. learning to adapt to new threat vectors with few training exercises. These achievements and the reinforcement learning and the PPO algorithm all lead to the realization of the latter, respectively, as a basis of adaptive high-performance zero-day attack detection.

5.6.2 Implications for Cybersecurity

This outcome is an indication of a paradigm shift in the area of cybersecurity that is being utilized. Statically fixed signatures are being superseded with adaptive and smart threat. detection frameworks. The framework is through the use of PPO-based reinforcement learning agents. empowered to detect zero-day threats with a high success rate of 93.94 with a comparatively low. false positive rate = 2.21, and is directly related to the issue of detecting new threats without. saturating the analyst with false positives. It is so achievable through such additions. to work in enterprise environments: It is possible to have intelligent Security Operations Centers. automated detection and initial analysis; the incident response teams can receive prompt

and autonomous zero-day threat detection to accelerate the containment and mitigation; and general risk is reduced by proactive real-time protection against advanced, persistent threats. The flexibility of the model is not confined to its incorporation as the basis. because of intrusion detection systems, yet also because of real-time network traffic analysis, as well as, too. protecting hosts endpoints, delivering a proactive, scalable cybersecurity strategy. Moreover, proper automated detection power and smart threat indication become easy. organizational resources, which force the discipline into the direction of truly intelligent, proactive. cyber defense.

5.6.3 Technical Implications

The excellent performance of PPO compared to both DQN and SAC in the field of cybersecurity indicates that there are other inherent merits of policy-based learning in detecting the emerging threats. PPO and other policy gradient approaches are rather effective in formulating their detection policies to novel and changed patterns of attack through continuous adaptation between exploration and exploitation by updating their clipped policies to stabilize learning and exploration of the unfamiliar threat. Another benefit of PPO is the actor-critic framework on which it is built, which further increases the sequential decision-making performance that allows addressing the details and proposing the explainable characterization that can be valuable to safety analysts. On that note, the discovery of the peculiarities due to the intuitiveness of the threshold-based features and the capability to take advantage of the high-dimensional features of a network offer flexible and adaptable boundaries of identification and learning, which highlights the practical application of reinforcement learning in the process of supporting flexible and current intrusion detection systems. Clipped updates enable the policy-based learning of PPO to keep adapting to the changing threats and balancing the explorations with exploitation in a stabilized environment. This flexibility and stability is provided at an additional interpretability of the model and the choices made through PPO can be understood by security analysts. Architecturally, the actor-critic designs are observed to definitely favor the sequential network security decision-making, whose threshold-based classifications possess an intuitive, adjustable boundary to decisions. Generally, reinforcement learning methods like PPO have achieved high-dimensional network feature space complexities to effectively detect anomalies and possibly useful and explainable deployment to assist advanced cybersecurity activities.

5.6.4 Limitations and Challenges

There have been some limitations and shortcomings of the system which are worth stating. The quality and representativeness of training sets significantly influence its performance and the detection precision may be affected for other attacks or unexpected network scenarios not addressed. Hyperparameter sensitivity, especially the learning rate, makes deployment difficult, requiring skilled expertise for fine-tuning, with a high degree of impact on performance of 23.3%. Though PPO is clearer than a deep Q-network, decision-making is somewhat cryptic, which restricts its usability.

ity on those occasions when it is desirable that the security decisions remain clear. Moreover, resilience to adversarial attacks specifically designed to avoid detection was not thoroughly tested and it remains unknown whether more advanced attackers may use the weaknesses of models to avoid or mislead the detection system.

5.6.5 Practical Deployment Considerations

There are several fundamental considerations that should be addressed in deployment phase of the proposed model. New security infrastructures need to be carefully designed at the API level to be compatible with the legacy security infrastructure, requiring standard data formats. Scalability should also be considered to meet the needs of high-throughput environments, which may require distributed deployment architectures capable of effectively managing the high amount of network traffic. Its ability to learn continuously necessitates the adoption of powerful update procedures, such as thoroughly tested update and rollback processes to ensure reliability throughout improvement cycles. Operationally, although the false positive rate is low (2.21), sufficient false positive management and feedback strategies are necessary to avoid analyst fatigue. Ongoing performance tracking should be implemented to ensure the same accuracy in detecting threats over time, and automated threat detection should be seamlessly integrated with the existing incident response processes to facilitate rapid containment and mitigation of emerging threats.

5.6.6 Future Research Directions

This research can be used to make several future changes to the model based on the tasks of the advanced capabilities such as less operation time and false positive. Technically, the implementation of multi-agent systems comprising collaborative agents in the shape of PPO. Transfer learning and domain adaptation methods would be required to enable generalizability to the various network settings, and adversarial training will be required to enable resistance to advanced evasion strategies. It will also need development of explainable AI models, which will facilitate the improvement of interpretability and confidence in computer-generated cybersecurity decisions. The practical implications will be expanded into the broader domains of using them to resource-constrained internet of things, multi-tenant clouds, and smartphones. The method-wise, the design of the online learning functionality that would allow adaptive learning not necessitating the entire group to be re-trained in a full-scale scenario and the use of federated learning to offer privacy-resistant inter-organizational partnerships would be valuable investments. Lastly, network, host and user behavior based multi-modal detection may support the holistic detection of threats that will make holistic cybersecurity solutions more robust.

5.6.7 Contribution Significance

The study has important contributions that present a novel system of zero-day attack detection via reinforcement learning. It has an impressive performance with the PPO. It offers a new perspective of methodology that includes a large amount of ablation study as a part of RL-based cybersecurity implementation. It is the first paper to provide a detailed comparative analysis of the most popular RL schemes-PPO, DQN, and SAC in the cyberspace. Practically, the system can be applied to the industry with features of performance that are viable to production such as real-time functionality. Moreover, it is more cost-effective to the level of effective detection and computational effectiveness, and it can be extensively applied to next-generation autonomous cybersecurity system.

Chapter 6

Conclusion

6.1 Summary of Findings

- Demonstrated that Proximal Policy Optimization (PPO), combined with Siamese cross validation delivers an effective and adaptable intrusion detection system by associating anti-anomaly patterns.
- Successfully confirms cross-dataset generalization of CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024 datasets.
- Exhibited remarkable training accuracy, unseen accuracy and zero-day accuracy.
- Experimentally identified learning rate as the most sensitive hyperparameter.
- Latency was 0.50 ms/ sample and false positive rate while not remarkable could be improved with further tuning.
- Comparative analysis revealed overall best sample efficacy and faster convergence, reaching 90% accuracy by episode 42 for PPO.

6.2 Contributions to the Field

The research also adds to the intersections between artificial intelligence and cybersecurity in a few significant respects. First, we introduce a new three-layer detection architecture that combines unsupervised anomaly detection (PCA and Autoencoder) with cross-dataset similarity analysis (Siamese network) and adaptive response (reinforcement learning); based on literature review that was done for this research, this type of approach has not been previously explored on zero-day attack detection. Our comprehensive ablation work provides the cybersecurity research community with empirical data on the hyperparameter sensitivity rankings

and the optimal decisions in the hyperparameter settings of PPO-based intrusion detection models, which has been missing in the deployment-oriented literature on RL. The cross-dataset anomaly filtering distance measure based optimization model with three similarities, i.e., Euclidean, Manhattan and Cosine measures, is a reproducible research approach to find uncommon in cross-network spaces dangerous scenarios with high consequences. We demonstrate that, as a methodological contribution, policy-based reinforcement learning can greatly outperform value-based (DQN) and complex actor-critic (SAC) versions of the system, with only learning rate optimization, and without consuming a costly memory footprint (23.3% better performance, 203.56MB of memory). The results of our research establish new performance goals in the zero-day detection systems under RL, the accuracy rates (99.07% unseen, 99.69% ROC AUC) are below the existing literature and deployed capabilities of cybersecurity activities in enterprises, and the gap between research and practice on cybersecurity has been bridged.

6.3 Recommendations for Future Work

There are various significant directions that this framework can be extended in the future so as to make it even more robust, scalable and versatile. A key step is adversarial robustness testing, which would test the systems resilience to adversarial attacks that are specifically designed to bypass detection boundaries. The experiments are necessary before rolling out the framework to mission-critical or safety-intensive environments. Furthermore, advancing multi-agent reinforcement learning (MARL) where distributed PPO or SAC agents engage in the protection of multiple layers of network infrastructure (e.g., IoT, cloud, and endpoint architecture) would enable a broader range of scalability and responsiveness in the large enterprise context. Attention-based visualization and SHAP value analysis will enhance the interpretability of the policy formulation. Incorporation of interoperability will also help the system comply with the regulation in highly regulated sectors. Furthermore, by incorporating transfer learning and domain adaptation, the model will be able to extrapolate to a wide range of network ecosystems, including industrial control systems, mobile applications, or blockchain networks, without bearing retraining costs and time to deploy. By introducing constant and online learning systems, it is possible to keep the IDS current with the changing zero-day threats and avoid catastrophic forgetting when updating it in an incremental manner. The second avenue of promise is federated learning, whereby multiple organizations can cooperatively use a single model without exposing sensitive information to fulfill privacy and data sovereignty requirements. Lastly, the trend of evolving to multi-modal threat intelligence fusion, in which network traffic data, host-based telemetry, and global threat feeds are combined and analyzed collectively, may result in the creation of fully autonomous, self-reinforcing cybersecurity systems, with the ability to adapt in real-time and continuously improve themselves.

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